Chiang Mai University

# UndefinedAC + HomuraAkemi + Intercontinental Ballistic Missile CMU

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# Contest (1)

template.cpp

80 lines

```
#include<bits/stdc++.h>
using namespace std;
void __print(int x) {cerr << x;}
void __print(long x) {cerr << x;}
void __print(long long x) {cerr << x;}
void __print(unsigned x) {cerr << x;}
void __print(unsigned long x) {cerr << x;}
void __print(unsigned long long x) {cerr << x;}
void __print(float x) {cerr << x;}
void __print(double x) {cerr << x;}
void __print(long double x) {cerr << x;}
void __print(char x) {cerr << '\'' << x << '\'';}
void __print(const char *x) {cerr << '\"' << x << '\"';}
void __print(const string &x) {cerr << '\"' << x << '\"';}
void __print(bool x) {cerr << (x ? "true" : "false");}

template<typename T, typename V>
void __print(const pair<T, V> &x);
template<typename T>
void __print(const T &x) {int f = 0; cerr << '{'; for (auto &i:
     x) cerr << (f++ ? ", " : ""), __print(i); cerr << "}";}
template<typename T, typename V>
void __print(const pair<T, V> &x) {cerr << '{'; __print(x.first
    ); cerr << ", "; __print(x.second); cerr << '}';}
void _print() {cerr << "]\n";}
template <typename T, typename... V>
void _print(T t, V... v) {__print(t); if (sizeof...(v)) cerr <<
     ", "; _print(v...);}
//#ifdef DEBUG
#define dbg(x...) cerr << "\e[91m"<<__func__<<":"<<__LINE__<<"
    [" << #x << "] = ["; _print(x); cerr << "\e[39m" << endl;
//#else
//#define dbg(x...)
//#endif

typedef long long ll;
typedef long double ld;
typedef complex<ld> cd;

typedef pair<int, int> pi;
typedef pair<ll,ll> pl;
typedef pair<ld,ld> pd;

typedef vector<int> vi;
typedef vector<ld> vd;
typedef vector<ll> vl;
typedef vector<pi> vpi;
typedef vector<pl> vpl;
typedef vector<cd> vcd;

//template<class T> using pq = priority_queue<T>;
//template<class T> using pqg = priority_queue<T, vector<T>,
    greater<T>>;
```

```
#define rep(i, a) for(int i=0;i<a;++i)
#define FOR(i, a, b) for (int i=a; i<(b); i++)
#define F0R(i, a) for (int i=0; i<(a); i++)
#define FORd(i,a,b) for (int i = (b)-1; i >= a; i--)
#define F0Rd(i,a) for (int i = (a)-1; i >= 0; i--)
#define trav(a,x) for (auto& a : x)
#define uid(a, b) uniform_int_distribution<long long>(a, b)(rng
    )

#define sz(x) (int)(x).size()
#define mp make_pair
#define pb push_back
//#define f first
//#define s second
#define lb lower_bound
#define ub upper_bound
#define all(x) x.begin(), x.end()
#define ins insert

template<class T> bool ckmin(T& a, const T& b) { return b < a ?
     a = b, 1 : 0; }
template<class T> bool ckmax(T& a, const T& b) { return a < b ?
     a = b, 1 : 0; }

mt19937 rng(chrono::steady_clock::now().time_since_epoch().
    count());

const char nl = '\n';
const int N = 2e5+3;
const int INF = INT_MAX;
const long long LINF = LLONG_MAX;

int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
}
```

.vimrc

21 lines

```
"General editor settings
set tabstop=4
set nocompatible
set shiftwidth=4
set expandtab
set autoindent
set smartindent
set ruler
set showcmd
set incsearch
set shellslash
set number
set relativenumber
set cino+=L0

"keybindings for { completion, "jk" for escape, ctrl-a to
    select all
inoremap {<CR>  {<CR>}<Esc>O
inoremap {}     {}
imap jk         <Esc>
map <C-a> <esc>ggVG<CR>
set belloff=all
```

.bashrc

1 lines

```
export PATH=$PATH:~/scripts/
```

build.sh

1 lines

```
g++ -static -DLOCAL -lm -s -x c++ -Wall -Wextra -O2 -std=c++17
    -o $1 $1.cpp
```

stress.sh

22 lines

```
#!/usr/bin/env bash

for ((testNum=0;testNum<$4;testNum++))
do
    ./$3 > input
    ./$2 < input > outSlow
    ./$1 < input > outWrong
    H1=`md5sum outWrong`
    H2=`md5sum outSlow`
    if !(cmp -s "outWrong" "outSlow")
    then
        echo "Error found!"
        echo "Input:"
        cat input
        echo "Wrong Output:"
        cat outWrong
        echo "Slow Output:"
        cat outSlow
        exit
    fi
done
echo Passed $4 tests
```

# Tanya's Library (2)

## 2.1 Template & Utilities

pragma.h

**Description:** random pragma magic that I don't fucking know but ocassionaly solve my stupid bad implementation code

1021ce, 7 lines

```
//some one on cf told us that these are mostly enough
#pragma GCC target ("avx2")
#pragma GCC optimize ("O3")
#pragma GCC optimize ("unroll-loops")
//just in case have this one
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,popcnt,lzcnt")
```

## 2.2 Basic & Combinatorics

binpow.h

**Description:** TODO

d4debd, 11 lines

```
long long binpow(long long a, int n){
    long long res = 1;
    while(n>0){
        if(n&1){
            res = res * a;
        }
        a = a * a;
        n>>=1;
    }
    return res;
}
```

bigmod.h

**Description:** TODO

91cb0a, 11 lines

```
long long bigmod(long long a, int n){
    long long res = 1;
    while(n>0){
        if(n&1){
            res = res * a % MOD;
        }
        a = a * a % MOD;
        n>>=1;
```

```
    }
    return res;
}
```

## binomialCoefficients.h

**Description:** TODO

3f7261, 10 lines

```
//more practical to precal C[n][r] with pascal triangle...
//...if constrain is O(n^2) able
long long C(int n, int k) {
    if(k==0)return 1;
    else if(k<0 || n<k)return 0;
    double res = 1;
    for (int i = 1; i <= k; ++i)
        res = res * (n - k + i) / i;
    return (long long)(res + 0.01);
}
```

## matrix.h

**Description:** Nothing to says

5da844, 33 lines

```
struct Matrix {
    vector<vector<long long>> a;
    int rows, cols;

    Matrix(int r, int c, int val) : rows(r), cols(c), a(r,
         vector<long long>(c, val)) {}

    Matrix operator *(const Matrix& other) {
        Matrix product(rows, other.cols, 0);

        for (int i = 0; i < rows; ++i) {
            for (int j = 0; j < other.cols; ++j) {
                for (int k = 0; k < cols; ++k) {
                    product.a[i][j] += a[i][k] * other.a[k][j]
                         % INF;
                    product.a[i][j] %= INF;
                }
            }
        }
        return product;
    }
};

Matrix expo_power(Matrix a, long long n, int size){//note n=1
     is equal do nothing,n=2is do once
    Matrix product(size,size,0);
    for(int i=0;i<size;++i)product.a[i][i]=1;
    while(n>0){
        if(n&1){
            product = product * a;
        }
        a = a * a;
        n>>=1;
    }
    return product;
}
```

## mobius-function.h

**Description:** TODO

c6a8f4, 10 lines

```
int mob[N+1];
void mobius() {
  mob[1] = 1;
  for (int i = 2; i <= N; i++){
    mob[i]--;
    for (int j = i + i; j <= N; j += i) {
      mob[j] -= mob[i];
    }
  }
}
```

## div-ceil-floor.h

**Description:** to stop c++ fucked up negative case of ceil, floor

2834e8, 9 lines

```
long long div_floor(__int128 a, long long b){
    __int128 q = a / b;
    __int128 r = a % b;
    if (r != 0 && ((a < 0) != (b < 0))) --q;
    return (long long)q;
}
long long div_ceil(__int128 a, long long b){
    return -div_floor(-a, b);
}
```

## factoradix.h

**Description:** TODO

<ext/pb_ds/assoc_container.hpp>, <ext/pb_ds/tree_policy.hpp>

e835c3, 63 lines

```
typedef __gnu_pbds::tree<int, __gnu_pbds::null_type, less<int>,
     __gnu_pbds::rb_tree_tag, __gnu_pbds::
     tree_order_statistics_node_update> ordered_set;

ordered_set st;
//st.order_of_key(x) - find # of elements in st stricly less
     than x
//st.size() - size of st
//st.find_by_order(x) - return iterator to the x-th largest
     element
//st.clear() - clear container

vector<int> factoradix_to_permutation(int n, vector<int> &
     factoradix){
    st.clear(); for(int i=1;i<=n;++i) st.ins(i);
    int cl = 0;
    while((int)factoradix.size() < n){
        factoradix.pb(0); ++cl;
    }

    vector<int> res;
    for(int i=(int)factoradix.size()-1;i>=0;--i){
        auto it = st.find_by_order(factoradix[i]);
        res.pb(*it);
        st.erase(it);
    }

    while(cl--) factoradix.pop_back();

    return res;
}

vector<int> permutation_to_factoradix(int n, vector<int> &
     permutation){
    st.clear(); for(int i=1;i<=n;++i) st.ins(i);
    vector<int> res;
    for(int i=0;i<(int)permutation.size();++i){
        int cnt = st.order_of_key(permutation[i]);
        res.pb(cnt);
        auto it = st.find_by_order(cnt); st.erase(it);
    }
    reverse(all(res));
    while(!res.empty() and res.back() == 0)res.pop_back();

    return res;
}

vector<int> decimal_to_factoradix(ll n){
    vector<int> res;
    int d = 1;
    while(n > 0){
        res.pb(n % d);
        n/=d; ++d;
    }
    return res;
}

ll factoradix_to_decimal(vector<int> &cur){
    ll res = 0;
    ll d = 1;
    for(int i=0;i<(int)cur.size();++i){
        res += d*cur[i];
        d*=(i+1);
    }
    return res;
}

//TODO using 0 base indexing, i.e. your 1st lexicographically
     permutation now is start at 0th
```

## gauss-jordan-elimination.h

**Description:** TODO

d847fe, 48 lines

```
const double EPS = 1e-9;
const int INF = 2; // it doesn't actually have to be infinity
     or a big number

int gauss (vector < vector<double> > a, vector<double> & ans) {
    int n = (int) a.size();
    int m = (int) a[0].size() - 1;

    vector<int> where (m, -1);
    for (int col=0, row=0; col<m && row<n; ++col) {
        int sel = row;
        for (int i=row; i<n; ++i)
            if (abs (a[i][col]) > abs (a[sel][col]))
                sel = i;

        if (abs (a[sel][col]) < EPS)
            continue;

        for (int i=col; i<=m; ++i)
            swap (a[sel][i], a[row][i]);
        where[col] = row;

        for (int i=0; i<n; ++i)
            if (i != row) {
                double c = a[i][col] / a[row][col];
                for (int j=col; j<=m; ++j)
                    a[i][j] -= a[row][j] * c;
            }
        ++row;
    }

    ans.assign (m, 0);
    for (int i=0; i<m; ++i)
        if (where[i] != -1)
            ans[i] = a[where[i]][m] / a[where[i]][i];

    for (int i=0; i<n; ++i) {
        double sum = 0;
        for (int j=0; j<m; ++j)
            sum += ans[j] * a[i][j];
        if (abs (sum - a[i][m]) > EPS)
            return 0; // contradiction means no solution
    }

    for (int i=0; i<m; ++i)
        if (where[i] == -1)
```

```
        return INF; //infinite solution
    return 1; //unique solution
}
```

## 2.3 Primes & Factorization

### sieve.h

**Description:** TODO

abb3a3, 22 lines

```
//construct is O(sqrt(n))
//can use to find all prime factor of a number in O(log n)
const int M = 2e5+1;
vector<bool> is_prime(M+1, true);
void sieve(){
    is_prime[0] = is_prime[1] = false;
    for (int i = 2; i * i <= M; i++) {
        if (is_prime[i]) {
            for (int j = i * i; j <= M; j += i)
                is_prime[j] = false;
        }
    }
    /* log n sieve (use sieve to find all prime factors in O(
        log n) )
    for (int i = 2; i <= M; i++)is_prime[i] = i;
    for (int i = 2; i * i <= M; i++) {
        if (is_prime[i] == i) {
            for (int j = i * i; j <= M; j += i)
                is_prime[j] = i;
        }
    }
    */
}
```

### linear-sieve.h

**Description:** TODO

d893f1, 18 lines

```
const int N = 10000000;
vector<int> lp(N+1);//lp[i] = minimum prime that divide i
vector<int> pr;

void sieve(){
    for (int i=2; i <= N; ++i) {
        if (lp[i] == 0) {
            lp[i] = i;
            pr.push_back(i);
        }
        for (int j = 0; i * pr[j] <= N; ++j) {
            lp[i * pr[j]] = pr[j];
            if (pr[j] == lp[i]) {
                break;
            }
        }
    }
}
```

### phi-sieve.h

**Description:** TODO

b4e4ad, 15 lines

```
//set value for M
const int M = 1e6;
vector<int> phi(M+1);

void phi_1_to_n() {
    for (int i = 0; i <= M; i++)
        phi[i] = i;

    for (int i = 2; i <= M; i++) {
        if (phi[i] == i) {
            for (int j = i; j <= M; j += i)
                phi[j] -= phi[j] / i;
        }
    }
}
```

### phi-function.h

**Description:** TODO

fb8d57, 14 lines

```
//O(sqrt n)
int phi(int n) {
    int result = n;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0)
                n /= i;
            result -= result / i;
        }
    }
    if (n > 1)
        result -= result / n;
    return result;
}
```

### mobius-function.h

**Description:** TODO

c6a8f4, 10 lines

```
int mob[N+1];
void mobius() {
  mob[1] = 1;
  for (int i = 2; i <= N; i++){
    mob[i]--;
    for (int j = i + i; j <= N; j += i) {
      mob[j] -= mob[i];
    }
  }
}
```

### miller-rabin.h

**Description:** TODO

980e40, 52 lines

```
//randomize primality test O(logn^2)
using u64 = uint64_t;
using u128 = __uint128_t;

u64 bigmod(u64 base, u64 e, u64 mod){
    u64 result = 1;
    base %= mod;
    while(e){
        if(e & 1)
            result = (u128)result * base % mod;
        base = (u128)base * base % mod;
        e >>= 1;
    }
    return result;
}

bool check_composite(u64 n, u64 a, u64 d, int s){
    u64 x = bigmod(a, d, n);
    if(x == 1 || x == n - 1)
        return false;
    for(int r = 1; r < s; r++){
        x = (u128)x * x % n;
        if(x == n - 1)
            return false;
    }
    return true;
}

//random number generator temporalily disable because my
    template already has it
//mt19937 rng(chrono::steady_clock::now().time_since_epoch().
    count());
long long rnd(long long x, long long y) {
  return uniform_int_distribution<long long>(x, y)(rng);
}

bool MillerRabin(u64 n, int iter=5){ // return true if n is
    probably prime, else return false.
    if(n < 4){
        return n == 2 || n == 3;
    }
    int s = 0;
    u64 d = n-1;
    while((d & 1) == 0){
        d >>= 1;
        s++;
    }

    for(int i = 0; i < iter; i++){
        long long a = rnd(2, n-2);
        if(check_composite(n, a, d, s))
            return false;
    }
    return true;
}
```

### pollard-rho.h

**Description:** TODO

bc7c06, 20 lines

```
//randomize factorization in O(n^1/4 * logn)
long long mult(long long a, long long b, long long mod) {
    return (__int128)a * b % mod;
}

long long f(long long x, long long c, long long mod) {
    return (mult(x, x, mod) + c) % mod;
}

long long rho(long long n, long long x0=2, long long c=1) {
    long long x = x0, y = x0, g = 1;
    while (g == 1) {
        x = f(x, c, n);
        y = f(f(y, c, n), c, n);
        g = __gcd(abs(x - y), n);
    }

    if(g == n) return rho(n, uid(2, n-1), uid(-2, 2)); //uid is
        uniform long
    else return g;
}
```

### discrete-log.h

**Description:** TODO

f7650d, 33 lines

```
// Returns minimum x for which a ^ x % m = b % m.
int solve(int a, int b, int m) {
    a %= m, b %= m;
    int k = 1, add = 0, g;
    while ((g = __gcd(a, m)) > 1) {
        if (b == k)
            return add;
        if (b % g)
            return -1;
        b /= g, m /= g, ++add;
        k = (k * 1ll * a / g) % m;
    }

    int n = sqrt(m) + 1;
    int an = 1;
```

```
    for (int i = 0; i < n; ++i)
        an = (an * 1ll * a) % m;

    map<int, int> vals;
    for (int q = 0, cur = b; q <= n; ++q) {
        vals[cur] = q;
        cur = (cur * 1ll * a) % m;
    }

    for (int p = 1, cur = k; p <= n; ++p) {
        cur = (cur * 1ll * an) % m;
        if (vals.count(cur)) {
            int ans = n * p - vals[cur] + add;
            return ans;
        }
    }
    return -1;
}
```

## primitive-root.h

**Description:** TODO

c6d472, 34 lines

```
/*
    The following code assumes that the modulo p is a prime
        number.
    To make it works for any value of p, we must add
        calculation of phi(p)
*/
int powmod (int a, int b, int p) {
    int res = 1;
    while (b)
        if (b & 1)
            res = int (res * 1ll * a % p),  --b;
        else
            a = int (a * 1ll * a % p),  b >>= 1;
    return res;
}

int generator (int p) {
    vector<int> fact;
    int phi = p-1,  n = phi; // cal phi p by assuming it's
        prime
    for (int i=2; i*i<=n; ++i)
        if (n % i == 0) {
            fact.push_back (i);
            while (n % i == 0)
                n /= i;
        }
    if (n > 1)
        fact.push_back (n);

    for (int res=2; res<=p; ++res) {
        bool ok = true;
        for (size_t i=0; i<fact.size() && ok; ++i)
            ok &= powmod (res, phi / fact[i], p) != 1;
        if (ok)  return res;
    }
    return -1;
}
```

## crt.h

**Description:** TODO

efd085, 51 lines

```
class CRT { // ChineseRemainderTheorem
    typedef long long vlong;
    typedef pair<vlong,vlong> pll;
    vector<pll> equations;

public:
    void clear() {
        equations.clear();
    }
    void addEquation( vlong r, vlong m ) {
        equations.push_back({r, m});
    }

    void ext_gcd(vlong a, vlong b, vlong& x, vlong& y) {
        if (b == 0) {
            x = 1;
            y = 0;
            return;
        }
        vlong x1, y1;
        ext_gcd(b, a % b, x1, y1);
        x = y1;
        y = x1 - y1 * (a / b);
    }

    pll solve() {
        if (equations.size() == 0) return {-1,-1};

        vlong a1 = equations[0].first;
        vlong m1 = equations[0].second;
        a1 %= m1;
        for ( int i = 1; i < (int)equations.size(); i++ ) {
            vlong a2 = equations[i].first;
            vlong m2 = equations[i].second;

            vlong g = __gcd(m1, m2);
            if ( a1 % g != a2 % g ) return {-1,-1};
            vlong p, q;
            ext_gcd(m1/g, m2/g, p, q);

            vlong mod = m1 / g * m2;
            p = (p%mod+mod)%mod, q = (q%mod+mod)%mod;
            vlong x = ( (__int128)a1 * (m2/g) % mod *q % mod +
                 (__int128)a2 * (m1/g) % mod * p % mod ) % mod;
            a1 = x;
            if ( a1 < 0 ) a1 += mod;
            m1 = mod;
        }
        return {a1, m1};
    }
};
```

## diophantine.h

**Description:** TODO

4fcad2, 89 lines

```
ll gcd(ll a, ll b, ll& x, ll& y) { //gcd extended
    if (b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    ll x1, y1;
    ll d = gcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

bool find_any_solution(ll a, ll b, ll c, ll &x0, ll &y0, ll &g)
    {
    g = gcd(abs(a), abs(b), x0, y0);
    if (c % g) {
        return false;
    }
    x0 *= c / g;
    y0 *= c / g;
    if (a < 0) x0 = -x0;
    if (b < 0) y0 = -y0;
    return true;
}

void shift_solution(ll & x, ll & y, ll a, ll b, ll cnt) { //
    make sure that a,b coprime
    x += cnt * b;
    y -= cnt * a;
}

ll find_all_solutions(ll a, ll b, ll c, ll minx, ll maxx, ll
    miny, ll maxy) {
    ll x, y, g;
    if (!find_any_solution(a, b, c, x, y, g))
        return 0;
    a /= g;
    b /= g;

    ll sign_a = a > 0 ? +1 : -1;
    ll sign_b = b > 0 ? +1 : -1;

    shift_solution(x, y, a, b, (minx - x) / b);
    if (x < minx)
        shift_solution(x, y, a, b, sign_b);
    if (x > maxx)
        return 0;
    ll lx1 = x;

    shift_solution(x, y, a, b, (maxx - x) / b);
    if (x > maxx)
        shift_solution(x, y, a, b, -sign_b);
    ll rx1 = x;

    shift_solution(x, y, a, b, -(miny - y) / a);
    if (y < miny)
        shift_solution(x, y, a, b, -sign_a);
    if (y > maxy)
        return 0;
    ll lx2 = x;

    shift_solution(x, y, a, b, -(maxy - y) / a);
    if (y > maxy)
        shift_solution(x, y, a, b, sign_a);
    ll rx2 = x;

    if (lx2 > rx2)
        swap(lx2, rx2);

    ll lx = max(lx1, lx2);
    ll rx = min(rx1, rx2);
    if (lx > rx)
        return 0;

    return (rx - lx) / abs(b) + 1;
}
//number of solution of ax + by = c in general
ll number_of_solution(ll a, ll b, ll c, ll x1, ll x2, ll y1, ll
    y2){
    if(a == 0 and b == 0 and c == 0)return (x2-x1+1)*(y2-y1+1);
    else if(a == 0 and b == 0) return 0;
    else if(a == 0){
        if(c%b!=0 or y1>c/b or y2<c/b)return 0;
        else return (x2-x1+1);
    } else if(b == 0){
        if(c%a!=0 or x1>c/a or x2<c/a)return 0;
        else return (y2-y1+1);
```

```
    } else {
        return find_all_solutions(a, b, c, x1, x2, y1, y2);
    }
}
```

## 2.4 Arrays & Trees

### dsu.h

**Description:** TODO

7efb0c, 23 lines

```
//TODO: initialized parent[] and _rank[] array

int find_set(int v) {
    if (v == parent[v])
        return v;
    return parent[v] = find_set(parent[v]);
}

void make_set(int v) {
    parent[v] = v;
    _rank[v] = 1;
}

void union_sets(int a, int b) {
    a = find_set(a);
    b = find_set(b);
    if (a != b) {
        if (_rank[a] < _rank[b])
            swap(a, b);
        parent[b] = a;
        _rank[a]+=_rank[b];
    }
}
```

### fenwick-tree.h

**Description:** TODO

a9def9, 27 lines

```
//TODO: use 1 base indexing only!!! becareful
vector<long long>bit;

//range sum point update(k=new_val-old_val)
void add(int i, int k){//add k to ith data and it's parent
     range so on
    while(i<(int)bit.size()){
        bit[i]+=k;
        i+=i&-i;//add last set bit
    }
}

ll sum(int i){//culmulative sum to ith data
    ll sum=0;
    while(i>0){
        sum+=bit[i];
        i-=i&-i;
    }
    return sum;
}

void build_bit(vl &a){
    bit=a;
    for(int i=1;i<(int)bit.size();++i){
        int p=i+(i&-i);//index to parent
        if(p<(int)bit.size())bit[p]+=bit[i];
    }
}
```

### normalSegtree.h

**Description:** TODO

d1ac68, 30 lines

```
//TODO: use 0 base indexing, initialize tree[] array (
     preferably 4 times size of nodes array
vector<long long>tree;
void update(int node,int n_l,int n_r,int q_i,long long value){
    if(n_r<q_i || q_i<n_l)return;
    if(q_i==n_l && n_r==q_i){
        tree[node] = value;
        return;
    }
    int mid = (n_r+n_l)/2;
    update(2*node,n_l,mid,q_i,value);
    update(2*node+1,mid+1,n_r,q_i,value);
    tree[node] = tree[2*node] + tree[2*node+1];
}

long long f(int node,int n_l,int n_r,int q_l,int q_r){
    if(n_r<q_l || q_r<n_l)return 0;
    if(q_l<=n_l && n_r<=q_r)return tree[node];
    int mid = (n_l+n_r)/2;
    return f(2*node,n_l,mid,q_l,q_r) + f(2*node+1,mid+1,n_r,q_l
        ,q_r);
}

int build_tree(vi &a,int n){
    tree.clear();
    int m=n;
    while(__builtin_popcount(m)!=1)++m;
    tree.resize(2*m+10,0);
    for(int i=0;i<n;++i)tree[i+m]=a[i];
    for(int i=m-1;i>=1;--i)tree[i]=tree[2*i]+tree[2*i+1];
    return m;
}
```

### lazySegtree.h

**Description:** TODO

59e123, 50 lines

```
//TODO: use 0 base indexing
vector<long long> tree,lazy;
void update(int node,int n_l,int n_r,int q_l,int q_r,int value)
     {
    if(lazy[node]!=0){
        tree[node]+=(long long)(n_r-n_l+1)*lazy[node]; // for
             range + update
        if(n_l!=n_r){
            lazy[2*node]+=lazy[node];
            lazy[2*node+1]+=lazy[node];
        }
        lazy[node] = 0;
    }
    if(n_r<q_l || q_r<n_l)return;
    if(q_l<=n_l && n_r<=q_r){
        tree[node]+=(long long)(n_r-n_l+1)*value; // for range
             + update
        if(n_l!=n_r){
            lazy[2*node]+=value;
            lazy[2*node+1]+=value;
        }
        return;
    }
    int mid = (n_r+n_l)/2;
    update(2*node,n_l,mid,q_l,q_r,value);
    update(2*node+1,mid+1,n_r,q_l,q_r,value);
    tree[node] = tree[2*node] + tree[2*node+1];
}

long long f(int node,int n_l,int n_r,int q_l,int q_r){
    if(lazy[node]!=0){
        tree[node]+=(long long)(n_r-n_l+1)*lazy[node];
        if(n_l!=n_r){
            lazy[2*node] += lazy[node];
            lazy[2*node+1] += lazy[node];
        }
        lazy[node] = 0;
    }
    if(n_r<q_l || q_r<n_l)return 0;
    if(q_l<=n_l && n_r<=q_r)return tree[node];
    int mid = (n_l+n_r)/2;
    return f(2*node,n_l,mid,q_l,q_r) + f(2*node+1,mid+1,n_r,q_l
        ,q_r);
}

int build_tree(vi &a,int n){
    tree.clear(); lazy.clear();
    int m=n;
    while(__builtin_popcount(m)!=1)++m;
    tree.resize(2*m+10,0); lazy.resize(2*m+10,0);
    for(int i=0;i<n;++i)tree[i+m]=a[i];
    for(int i=m-1;i>=1;--i)tree[i]=tree[2*i]+tree[2*i+1];
    return m;
}
```

### persistent-segtree.h

**Description:** TODO

c4a26b, 38 lines

```
//Warning! This implementation is not garbage collected, so the
     tree nodes aren't deleted even if the instance of the
     segment tree is taken out of scope.
struct Node {
    ll val;
  Node *l, *r;

  Node(ll x) : val(x), l(nullptr), r(nullptr) {}
  Node(Node *ll, Node *rr) {
    l = ll, r = rr;
    val = 0;
    if (l) val += l->val;
    if (r) val += r->val;
  }
  Node(Node *cp) : val(cp->val), l(cp->l), r(cp->r) {}
};

int n, cnt = 1;
ll a[200001];
Node *roots[200001];

Node *build(int l = 1, int r = n) {
  if (l == r) return new Node(a[l]);
  int mid = (l + r) / 2;
  return new Node(build(l, mid), build(mid + 1, r));
}

Node *update(Node *node, int val, int pos, int l = 1, int r = n
    ) {
  if (l == r) return new Node(val);
  int mid = (l + r) / 2;
  if (pos > mid) return new Node(node->l, update(node->r, val,
      pos, mid + 1, r));
  else return new Node(update(node->l, val, pos, l, mid), node
      ->r);
}

ll query(Node *node, int a, int b, int l = 1, int r = n) {
  if (l > b || r < a) return 0;
  if (l >= a && r <= b) return node->val;
  int mid = (l + r) / 2;
  return query(node->l, a, b, l, mid) + query(node->r, a, b,
      mid + 1, r);
}
```

## merge-sort-tree.h

**Description:** TODO

<ext/pb_ds/assoc_container.hpp>, <ext/pb_ds/tree_policy.hpp>, <ext/pb_ds/assoc_container.hpp>, <ext/pb_ds/tree_policy.hpp> 2d0612, 94 lines

```
//merge sort tree with segment tree
typedef __gnu_pbds::tree<pair<int,int>, __gnu_pbds::null_type,
     less<pair<int,int>>, __gnu_pbds::rb_tree_tag, __gnu_pbds::
     tree_order_statistics_node_update> ordered_set;
ordered_set st;
//st.order_of_key({x,-1}) - find # of elements in st stricly
     less than x
//st.clear() - clear container

vector<ordered_set> mtree;

ordered_set merge(ordered_set &a, ordered_set &b){
    ordered_set result;
    for(auto&p:a){
        result.ins(p);
    }
    for(auto&p:b){
        result.ins(p);
    }
    return result;
}

void update(int node,int n_l,int n_r,int q_i,int id,int old_val
     ,int value){
    if(n_r<q_i || q_i<n_l)return;
    if(q_i==n_l && n_r==q_i){
        auto it=mtree[node].find({old_val,id});
        mtree[node].erase(it);
        mtree[node].ins({value,id});
        return;
    }
    int mid = (n_r+n_l)/2;
    update(2*node,n_l,mid,q_i,id,old_val,value);
    update(2*node+1,mid+1,n_r,q_i,id,old_val,value);
    auto it=mtree[node].find({old_val,id});
    if(it!=mtree[node].end()){
        mtree[node].erase(it);
        mtree[node].ins({value,id});
    }
}

int f(int node,int n_l,int n_r,int q_l,int q_r, int value){
    if(n_r<q_l || q_r<n_l)return 0;
    if(q_l<=n_l && n_r<=q_r){
        return mtree[node].order_of_key({value,-1});
    }
    int mid = (n_l+n_r)/2;
    return f(2*node,n_l,mid,q_l,q_r,value)+f(2*node+1,mid+1,n_r
         ,q_l,q_r,value);
}

void build_mtree(vi &a){
    int n=(int)a.size();
    int m=n; while(__builtin_popcount(m)!=1)++m;
    //for(int i=0;i<2*m;++i)mtree[i].clear();
    mtree.resize(2*m);
    for(int i=0;i<n;++i)mtree[i+m].ins({a[i],i});
    for(int i=m-1;i>=1;--i)mtree[i]=merge(mtree[2*i],mtree[2*i
         +1]);
}
//merge sort tree with fenwick tree(BIT) (4 times less space)

typedef __gnu_pbds::tree<pair<int,int>, __gnu_pbds::null_type,
     less<pair<int,int>>, __gnu_pbds::rb_tree_tag, __gnu_pbds::
     tree_order_statistics_node_update> ordered_set;
ordered_set st;
//st.order_of_key(x) - find # of elements in st stricly less
     than x

//TODO: use 1 base indexing
vector<ordered_set>bit;

void update(int i,int k,int old_value, int new_value){
    while(i<(int)bit.size()){
        auto it=bit[i].find({old_value,k});
        assert(it!=bit[i].end());
        if(it!=bit[i].end()){
            bit[i].erase(it);
        }
        bit[i].ins({new_value,k});
        i+=i&-i;//add last set bit
    }
}

int F(int i, int k){//culmulative sum to ith data
    int sum=0;
    while(i>0){
        sum+=bit[i].order_of_key({k,-1});
        i-=i&-i;
    }
    return sum;
}

void build_bit(vi &a){
    bit.resize((int)a.size());
    for(int i=1;i<(int)a.size();++i)bit[i].ins({a[i],i});
    for(int i=1;i<(int)bit.size();++i){
        int p=i+(i&-i);//index to parent
        if(p<(int)bit.size()){
            for(auto&x:bit[i])bit[p].ins(x);
        }
    }
}
```

## 2D-segtree.h

**Description:** TODO

ee2c74, 82 lines

```
// 2D segtree stored as tree[4*n][4*m]
vector<vector<ll>> tree;
int n, m; // This shit in global remember
vector<vector<ll>> A; // This shit in global too

void buildY(int vx, int lx, int rx, int vy, int ly, int ry) {
    if (ly == ry) {
        if (lx == rx) {
            tree[vx][vy] = A[lx][ly];
        } else {
            tree[vx][vy] = tree[vx*2][vy] + tree[vx*2+1][vy];
        }
    } else {
        int my = (ly + ry) >> 1;
        buildY(vx, lx, rx, vy*2,   ly,   my);
        buildY(vx, lx, rx, vy*2+1, my+1, ry);
        tree[vx][vy] = tree[vx][vy*2] + tree[vx][vy*2+1];
    }
}

void buildX(int vx = 1, int lx = 1, int rx = n) {
    if (lx != rx) {
        int mx = (lx+rx)>>1;
        buildX(vx*2,   lx,   mx);
        buildX(vx*2+1, mx+1, rx);
    }
    buildY(vx, lx, rx, 1, 1, m);
}

void updateY(int vx, int lx, int rx, int x,
             int vy, int ly, int ry, int y, ll val) {
    if (ly == ry) {
        if (lx == rx) {
            tree[vx][vy] = val;
        } else {
            tree[vx][vy] = tree[vx*2][vy] + tree[vx*2+1][vy];
        }
    } else {
        int my = (ly+ry)>>1;
        if (y <= my)
            updateY(vx,lx,rx,x, vy*2,   ly,   my, y, val);
        else
            updateY(vx,lx,rx,x, vy*2+1, my+1, ry, y, val);
        tree[vx][vy] = tree[vx][vy*2] + tree[vx][vy*2+1];
    }
}

void updateX(int vx, int lx, int rx, int x, int y, ll val) {
    if (lx != rx) {
        int mx = (lx+rx)>>1;
        if (x <= mx) updateX(vx*2,   lx,   mx, x, y, val);
        else         updateX(vx*2+1, mx+1, rx, x, y, val);
    }
    updateY(vx, lx, rx, x, 1, 1, m, y, val);
}

ll queryY(int vx, int vy, int ly, int ry, int y1, int y2) {
    if (y2 < ly || ry < y1) return 0;
    if (y1 <= ly && ry <= y2) return tree[vx][vy];
    int my = (ly+ry)>>1;
    return queryY(vx,vy*2,  ly,   my, y1,y2)
         + queryY(vx,vy*2+1,my+1, ry,  y1,y2);
}

ll queryX(int vx, int lx, int rx,
          int x1, int x2, int y1, int y2) {
    if (x2 < lx || rx < x1) return 0;
    if (x1 <= lx && rx <= x2)
        return queryY(vx, 1, 1, m, y1, y2);
    int mx = (lx+rx)>>1;
    return queryX(vx*2,   lx,   mx, x1,x2,y1,y2)
         + queryX(vx*2+1, mx+1, rx, x1,x2,y1,y2);
}

void build(int n, int m){
    tree.assign(4*(n+2), vector<ll>(4*(m+2), 0));
    buildX();
}

//using like this n,m = row,col dimention (1-base index)
//updateX(1,1,n,x,y,v);
//queryX(1,1,n,x1,x2,y1,y2)
```

## order-statistics-tree.h

**Description:** TODO

<ext/pb_ds/assoc_container.hpp>, <ext/pb_ds/tree_policy.hpp> 033b41, 7 lines

```
typedef __gnu_pbds::tree<int, __gnu_pbds::null_type, less<int>,
     __gnu_pbds::rb_tree_tag, __gnu_pbds::
     tree_order_statistics_node_update> ordered_set;

ordered_set st;
//st.order_of_key(x) - find # of elements in st stricly less
     than x
//st.size() - size of st
//st.find_by_order(x) - return iterator to the x-th largest
     element
```

```
//st.clear() - clear container
```

RMQ-1D.h

**Description:** TODO

e37d08, 25 lines

```
//general sparse table template(can be use for all other
     offline queries)
int rmq[N][20];

void build_rmq(vi &a){
    for(int j=0;j<20;++j){
        for(int i=0;i<(int)a.size();++i){
            if(j==0){
                rmq[i][0]=a[i];
            } else if(i+(1<<(j-1))<(int)a.size()){
                rmq[i][j]=min(rmq[i][j-1],rmq[i+(1<<(j-1))][j
                    -1]);
            }
        }
    }
}

int query(int l, int r){
    int i=l,sub_array_size=r-l+1, ans=INF;
    for(int j=0;j<30;++j){
        if((1<<j)&(sub_array_size)){
            ans=min(ans,rmq[i][j]);
            i+=(1<<j);
        }
    }
    return ans;
}
```

RMQ-2D.h

**Description:** TODO

0f2f45, 46 lines

```
//general template for 2D offline queries
int rmq[N][N][20][20]; //N need to be around 1e2

void build_rmq(vector<vi>& a){
    for(int j=0;j<20;++j){
        for(int k=0;k<20;++k){
            for(int r=0;r<(int)a.size();++r){
                for(int c=0;c<(int)a[0].size();++c){
                    if(j==0&&k==0)rmq[r][c][0][0]=a[r][c];
                    else {
                        if(j==0 && k>0 && (c+(1<<(k-1))<(int)a
                            [0].size() )){
                            rmq[r][c][j][k]=min(rmq[r][c][j][k
                                -1],
                            rmq[r][c+(1<<(k-1))][j][k-1]);
                        } else if(k==0 && j>0 && (r+(1<<(j-1))
                            <(int)a.size()) ){
                            rmq[r][c][j][k]=min(rmq[r][c][j-1][
                                k],
                            rmq[r+(1<<(j-1))][c][j-1][k]);
                        } else if( c+(1<<(k-1))<(int)a[0].size
                            ()
                        && r+(1<<(j-1))<(int)a.size() ){
                            rmq[r][c][j][k]=min({rmq[r][c][j
                                -1][k-1],
                            rmq[r][c+(1<<(k-1))][j-1][k-1],
                            rmq[r+(1<<(j-1))][c][j-1][k-1],
                            rmq[r+(1<<(j-1))][c+(1<<(k-1))][j
                                -1][k-1]});
                        }
                    }
                }
            }
        }
    }
}

int query(int r1,int c1,int r2,int c2){
    int x=r1,y=c1,h=r2-r1+1,w=c2-c1+1,ans=INF;
    for(int j=0;j<30;++j){
        if((1<<j)&h){
            for(int k=0;k<30;++k){
                if((1<<k)&w){
                    ans=min(ans,rmq[x][y][j][k]);
                    y+=(1<<k);
                }
            }
            y=c1;
            x+=(1<<j);
        }
    }
    return ans;
}
```

## 2.5 Hashing & Tables

gp-hash-table.h

**Description:** TODO

<ext/pb_ds/assoc_container.hpp> 3b295b, 7 lines

```
using namespace __gnu_pbds;

const int RANDOM = chrono::high_resolution_clock::now().
    time_since_epoch().count();
struct chash {
    int operator()(int x) const { return x ^ RANDOM; }
};
gp_hash_table<int, int, chash> table;
```

cht.h

**Description:** TODO

87ad47, 28 lines

```
struct line {
    long long m, c;
    long long eval(long long x) { return m * x + c; }
    long double intersectX(line l) { return (long double) (c -
        l.c) / (l.m - m); }
};

deque<line> dq;
dq.push_front({0, 0});//cant be put in global, remove this to
    local function
//if query ask for minimum remove this line after 1st insertion

//TODO NOTE***: maximum and minimum value exist in bot left
    most and rightmost of convex hull so do search on both l
    to r and r to l

//constructing hull from l to r, maintain correct hull at
    rightmost
/*  ***inserting line (maximum hull)
    line cur = line{...some m, ...some c}
    while(dq.size()>=2&&cur.intersectX(dq.back())
        <=cur.intersectX(dq[dq.size()-2]))dq.pop_back();
    dq.pb(cur);
*/

//constructing hull from r to l, maintain correct hull at
    leftmost
/*  inserting line (maximum hull)
    line cur = line{...some m, ...some c}
    while(dq.size()>=2&&cur.intersectX(dq[0])>=cur.intersectX(
        dq[1])){
        dq.pop_front();
    }
    dq.push_front(cur);
*/
```

## 2.6 Graph Algorithms

dijkstra.h

**Description:** TODO

3fd08c, 17 lines

```
//initialize dist, proc
    for (int i = 1; i <= m; i++) dist[i] = INF;
    dist[x] = 0;
    priority_queue<pair<ll, int>>q;
    q.push({0,x});
    while (!q.empty()) {
        int a = q.top().second; q.pop();
        if (proc[a]) continue;
        proc[a] = true; // this fucker i dont remember all the
            time
        for (auto u : adj[a]) {
            int b = u.first, w = u.second;
            if (dist[a]+w < dist[b]) {
                dist[b] = dist[a]+w;
                q.push({-dist[b],b});
            }
        }
    }
```

floyd-warshall-stp.h

**Description:** before k-th phase d[i][j] = stp len from i to j using only vertices from set 1,2,...k-1 as internal path, the beginning and end of path are not restricted by this property.

ab1dcf, 10 lines

```
//don't forget to initialize the base case of weight
//this work for negative weight without negative cycle
for (int k = 0; k < n; ++k) {
    for (int i = 0; i < n; ++i) {
        for (int j = 0; j < n; ++j) {
            if (d[i][k] < INF && d[k][j] < INF)
                d[i][j] = min(d[i][j], d[i][k] + d[k][j]);
        }
    }
}
```

kruskal-mst.h

**Description:** TODO

5dd25a, 37 lines

```
//kruskal
vector<tuple<int,int,int>>el;
int parent[N+1], _rank[N+1];

int find_set(int v) {
    if (v == parent[v])
        return v;
    return parent[v] = find_set(parent[v]);
}

void make_set(int v) {
    parent[v] = v;
    _rank[v] = 1;
}

void union_sets(int a, int b) {
    a = find_set(a);
    b = find_set(b);
    if (a != b) {
        if (_rank[a] < _rank[b])
            swap(a, b);
        parent[b] = a;
```

```
        _rank[a]+=_rank[b];
    }
}

ll mst(int n){
    sort(all(el)); for(int i=1;i<=n;++i)make_set(i);
    ll mst_cost=0;
    for(auto&t:el){
        int w=get<0>(t), a=get<1>(t), b=get<2>(t);
        if(find_set(a) != find_set(b)){
            mst_cost+=w, union_sets(a,b);
        }
    }
    return mst_cost;
}
```

prim-mst.h
**Description:** TODO
a70ceb, 24 lines

```
//*note: unlike kruskal, prim's algorithm assume all vertices
     belong in same graph
vector<pair<int,int>>adj[N+1];
bool taken[N+1];
priority_queue<pair<int,int>>PQ;

void process(int v){
    taken[v]=1;
    for(auto&p:adj[v]){
        int b=p.second, w=p.first;
        if(!taken[b])PQ.push({-w,-b});
    }
}

ll mst(){
    process(1);
    ll mst_cost=0;
    while(!PQ.empty()){
        pair<int,int> front=PQ.top(); PQ.pop();
        int u=-front.second, w=-front.first;
        if(!taken[u])
            mst_cost+=w, process(u);
    }
    return mst_cost;
}
```

dinic.h
**Description:** TODO
ce6220, 82 lines

```
//COPY FROM BENQ GITHUB I dont fucking know how this work
//General graphs: O(V^2E);
//Bipartite matching(unit caps): O(E sqrt2(V))
struct Dinic {
    using F = long long; // flow type
    struct Edge { int to; F flow, cap; };
    int N;
    vector<Edge> edges;
    vector<vector<int>> adj;
    vector<int> level;
    vector<vector<int>::iterator> it;

    // Initialize Dinic for a graph with _N vertices (0-indexed
        or 1-indexed)
    void init(int _N) {
        N = _N;
        adj.assign(N, {});
        edges.clear();
        level.resize(N);
        it.resize(N);
    }

    // Add edge u -> v with capacity 'cap', and reverse edge v
        -> u with capacity 'rev_cap'
    void addEdge(int u, int v, F cap, F rev_cap = 0) {
        // forward edge
        adj[u].push_back((int)edges.size());
        edges.push_back({v, 0, cap});
        // reverse edge
        adj[v].push_back((int)edges.size());
        edges.push_back({u, 0, rev_cap});
    }

    // Build level graph via BFS
    bool bfs(int s, int t) {
        fill(level.begin(), level.end(), -1);
        for (int i = 0; i < N; i++) it[i] = adj[i].begin();
        queue<int> q;
        level[s] = 0;
        q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int idx : adj[u]) {
                const Edge &e = edges[idx];
                if (level[e.to] < 0 && e.flow < e.cap) {
                    level[e.to] = level[u] + 1;
                    q.push(e.to);
                }
            }
        }
        return level[t] >= 0;
    }

    // DFS to send flow
    F dfs(int u, int t, F pushed) {
        if (u == t || pushed == 0) return pushed;
        for (; it[u] != adj[u].end(); ++it[u]) {
            int idx = *it[u];
            Edge &e = edges[idx];
            if (level[e.to] != level[u] + 1 || e.flow == e.cap)
                 continue;
            F tr = dfs(e.to, t, min(pushed, e.cap - e.flow));
            if (tr > 0) {
                e.flow += tr;
                edges[idx ^ 1].flow -= tr; // reverse edge
                return tr;
            }
        }
        return 0;
    }

    // Compute max flow from s to t
    F maxFlow(int s, int t) {
        F flow = 0;
        while (bfs(s, t)) {
            while (F pushed = dfs(s, t, numeric_limits<F>::max
                ())) {
                flow += pushed;
            }
        }
        return flow;
    }

    // After maxFlow, vertices reachable from s in residual
        graph have level >= 0.
    // Edges from reachable u to unreachable v form a min-cut.
};
```

eulerian-cycle.h
**Description:** TODO
87df80, 25 lines

```
//TODO: checking all node degrees is even ?
    //TODO: checking if all edges are connectedly reachable

    stack<int> st;
    vi res;
    st.push(1);
    while(!st.empty()){
        int v = st.top();
        if(deg[v] == 0){
            res.pb(v);
            st.pop();
        } else {
            int rm = -1; //must exist
            for(auto &b:adj[v]){
                rm = b;
                break;
            }
            adj[v].erase(rm); deg[v]--;
            adj[rm].erase(v); deg[rm]--;
            st.push(rm);
        }
    }
    //reverse the res if the edge is directed !! also changing
        the checking condition
    for(auto &x:res)cout << x << " ";
    cout << nl;
```

## 2.7 Graph Decompositions & Connectivity

articulation-points.h
**Description:** TODO
23f413, 37 lines

```
int n; // number of nodes
vector<vector<int>> adj; // adjacency list of graph

vector<bool> visited;
vector<int> tin, low;
int timer;

void dfs(int v, int p = -1) {
    visited[v] = true;
    tin[v] = low[v] = timer++;
    int children=0;
    for (int to : adj[v]) {
        if (to == p) continue;
        if (visited[to]) {
            low[v] = min(low[v], tin[to]);
        } else {
            dfs(to, v);
            low[v] = min(low[v], low[to]);
            if (low[to] >= tin[v] && p!=-1)
                IS_CUTPOINT(v);
            ++children;
        }
    }
    if(p == -1 && children > 1)
        IS_CUTPOINT(v);
}

void find_cutpoints() {
    timer = 0;
    visited.assign(n, false);
    tin.assign(n, -1);
    low.assign(n, -1);
    for (int i = 0; i < n; ++i) {
```

```
        if (!visited[i])
            dfs (i);
    }
}
```

## bridge-finding.h

**Description:** TODO

747a19, 30 lines

```
//NOTE**: not work for multigraph, I'm too lazy too fix for now
/*
    dfs-tree bridge finding algorithm
    let lvl[root] = 1 (root = 1)
*/
int root = 1;
vector<pair<int,int>> bridge;
int dp[N+1], lvl[N+1];

void dfs_tree(int a, int e){
    if(a == root)lvl[a] = 1;
    dp[a] = 0;
    for(auto &b:adj[a]){
        if(b == e)continue;

        if(lvl[b] == 0){
            lvl[b] = lvl[a] + 1;
            dfs_tree(b, a);
            dp[a] += dp[b];
        } else if(lvl[b] < lvl[a]){
            dp[a]++;
        } else if(lvl[b] > lvl[a]){
            dp[a]--;
        }
    }

    if(lvl[a] > 1 and dp[a] == 0){
        bridge.pb({min(a,e), max(a,e)});
    }
}
```

## tarjan-scc.h

**Description:** TODO

3e5381, 35 lines

```
//Trajan's algorithm
vi adj[N+1];
int id[N+1], low[N+1];
bool vst[N+1], in_stack[N+1];

stack<int>st;
vector<vi>scc;
vi comp;
int tin=0;
void dfs(int a){
    st.push(a);
    vst[a]=in_stack[a]=1;
    id[a]=low[a]=tin++;

    for(auto&b:adj[a]){
        if(!vst[b]){
            dfs(b);
            low[a]=min(low[a],low[b]); //Backtrack low-link
                 value
        } else if(in_stack[b]){
            //Backtrack low-link value
            low[a]=min(low[a],low[b]);
        }
    }
    if(id[a] == low[a]){ //pop scc
        while(!st.empty()){
            int v=st.top();
            comp.pb(v), st.pop();
            in_stack[v]=0;
            if(v==a)break; //stop when scc is found
        }
    }
    if(!comp.empty()){
        scc.pb(comp), comp.clear();
    }
}
```

## centroid-decomposition.h

**Description:** TODO

70d410, 32 lines

```
vector<int> adj[N];
bool is_removed[N];
int subtree_size[N];
int get_subtree_size(int node, int parent = -1) {
  subtree_size[node] = 1;
  for (int child : adj[node]) {
    if (child == parent || is_removed[child]) { continue; }
    subtree_size[node] += get_subtree_size(child, node);
  }
  return subtree_size[node];
}
int get_centroid(int node, int tree_size, int parent = -1) {
  for (int child : adj[node]) {
    if (child == parent || is_removed[child]) { continue; }
    if (subtree_size[child] * 2 > tree_size) {
      return get_centroid(child, tree_size, node);
    }
  }
  return node;
}
void build_centroid_decomp(int node = 0) {
  int centroid = get_centroid(node, get_subtree_size(node));
  is_removed[centroid] = true;

  // do something

  for (int child : adj[centroid]) {
    if (is_removed[child]) { continue; }
    build_centroid_decomp(child);
  }
}
```

## hld.h

**Description:** TODO

e4ffaf, 152 lines

```
/*hld*/
//TODO: initialize constant N
vector<int>adj[N];

void add_edges(int a, int b){
    adj[a].pb(b);
    adj[b].pb(a);
}

const int LOG = 20;
int timer = 0, seg_tree_size = 0;
int depth[N], parent[N], n_subtree[N], heavyChild[N], chain[N],
     SegTreePos[N];
int cost[N];
int up[N][LOG]; // 2^j-th ancestor of n

/*SegTree*/
//finding max edge query
ll tree[4*N];
void update(int node,int n_l,int n_r,int q_i,int value){
    if(n_r<q_i || q_i<n_l)return;
    if(q_i==n_l && n_r==q_i){
        tree[node] = value;
        return;
    }
    int mid = (n_r+n_l)/2;
    update(2*node,n_l,mid,q_i,value);
    update(2*node+1,mid+1,n_r,q_i,value);
    tree[node] = max(tree[2*node], tree[2*node+1]);
}

long long f(int node,int n_l,int n_r,int q_l,int q_r){
    if(n_r<q_l || q_r<n_l)return 0;
    if(q_l<=n_l && n_r<=q_r)return tree[node];
    int mid = (n_l+n_r)/2;
    return max(f(2*node,n_l,mid,q_l,q_r) , f(2*node+1,mid+1,n_r
         ,q_l,q_r));
}
/*SegTree*/

//preprecessing the arrays above
void dfs(int a,int e){
    chain[a] = a;
    n_subtree[a] = 1;
    int bestSon = -1, hmax = -1;
    for(auto b:adj[a]){
        if(b == e)continue;
        depth[b] = depth[a] + 1;
        parent[b] = a;
        up[b][0] = parent[b];
        for(int i=1;i<LOG;++i){
            up[b][i] = up[up[b][i-1]][i-1];
        }
        dfs(b,a);
        n_subtree[a] += n_subtree[b];
        if(n_subtree[b] > hmax){
            hmax = n_subtree[b], bestSon = b;
        }
    }
    heavyChild[a] = bestSon;
}

void computeHeavyChains(int a, int e){
    if(heavyChild[a] != -1){
        chain[heavyChild[a]] = chain[a];
    }
    for(auto b:adj[a]){
        if(b == e)continue;
        computeHeavyChains(b, a);
    }
}

void setHeavyChains(int a, int e){
    SegTreePos[a] = timer++;
    if(heavyChild[a] != -1){
        setHeavyChains(heavyChild[a], a);
        int w = 0;
        for(auto b:adj[a]){
            if(b == heavyChild[a]){
                w = 1;
                break;
            }
        }
        //setting cost of heavyChild[a] = weight of edge (a,
             heavyChild[a])
        cost[heavyChild[a]] = w; //change when tree is weighted
    }
    for(auto b:adj[a]){
        if(b == e || heavyChild[a] == b)continue;
        cost[b] = 1; //change when tree is weighted
```

```
        setHeavyChains(b ,a);
    }
}

void buildSegTree(int n_node){
    seg_tree_size = n_node;
    while(__builtin_popcount(seg_tree_size)!=1)++seg_tree_size;
    for(int i=0;i<n_node;++i){
        update(1,0,seg_tree_size-1,SegTreePos[i], cost[i]);
    }
    for(int i=seg_tree_size-1;i>=1;--i){
        tree[i] = max(tree[2*i], tree[2*i+1]);
    }
}

ll queryPath(int mother, int son){
    ll ans = LONG_MIN;
    while(chain[mother] != chain[son]){
        if(chain[son] == son){
            ans = max(ans,f(1,0,seg_tree_size-1,SegTreePos[son
                ],SegTreePos[son]));
        } else {
            ans = max(ans, f(1,0,seg_tree_size-1,SegTreePos[
                chain[son]],SegTreePos[son]));
        }
        son = parent[chain[son]];
    }

    ans = max(ans, f(1,0,seg_tree_size-1,SegTreePos[mother]+1,
        SegTreePos[son]));

    return ans;
}

int lca(int a, int b){
    if(depth[a]<depth[b])swap(a,b);
    int k = depth[a] - depth[b];
    for(int i=LOG-1;i>=0;--i){
        if(k &(1<<i)){
            a = up[a][i];
        }
    }

    if(a == b)return a;
    for(int i=LOG-1;i>=0;--i){
        if(up[a][i] != up[b][i]){
            a = up[a][i];
            b = up[b][i];
        }
    }
    return up[a][0];
}

ll query(int x, int y){
    int l = lca(x,y);
    return max( queryPath(l, x), queryPath(l, y)); //combine
        the answer

}

void hld_init(int n){
    dfs(0,-1);
    computeHeavyChains(0,-1);
    setHeavyChains(0,-1);
    buildSegTree(n);
}

/*hld*/
```

lca.h

**Description:** TODO

0ae36f, 36 lines

```
//TODO: initialize tree(adj list)
const int LOG = 20;
int dep[N], parent[N];
int up[N][LOG]; // 2^j-th ancestor of n

void dfs(int a,int e){
    for(auto b:adj[a]){
        if(b == e)continue;
        dep[b] = dep[a] + 1;
        parent[b] = a;
        up[b][0] = parent[b];
        for(int i=1;i<LOG;++i){
            up[b][i] = up[up[b][i-1]][i-1];
        }
        dfs(b,a);
    }
}

int lca(int a, int b){
    if(dep[a]<dep[b])swap(a,b);
    int k = dep[a] - dep[b];
    for(int i=LOG-1;i>=0;--i){
        if(k &(1<<i)){
            a = up[a][i];
        }
    }

    if(a == b)return a;
    for(int i=LOG-1;i>=0;--i){
        if(up[a][i] != up[b][i]){
            a = up[a][i];
            b = up[b][i];
        }
    }
    return up[a][0];
}
```

kth-ancestor.h

**Description:** TODO

3b29e6, 9 lines

```
int kth_ancestor(int node,int k){
    if(dep[node] < k)return -1;
    for(int i = 0;i < LOG; ++i){
        if(k & (1<<i)){
            node = up[node][i];
        }
    }
    return node;
}
```

tree-hashing.h

**Description:** TODO

ce0886, 22 lines

```
/*
    very acurate rooted tree hashing
    don't forget to check tree invariant such as tree size and
        # of centroid
*/
map<vector<int>, int> hasher;

int hashify(vector<int> x) {
    sort(x.begin(), x.end());
    if(!hasher[x]) {
        hasher[x] = hasher.size();
    }
    return hasher[x];
}

int _hash(int v, int e, vector<vector<int>>&adj) { // get a "
    hash" of v's subtree, e is parent vertex
    vector<int> children;
    for(int u: adj[v]) {
        if( u == e )continue;
        children.push_back(_hash(u, v, adj));
    }
    return hashify(children);
}
```

## 2.8 String Processing

AC-automaton.h

**Description:** TODO

a41621, 53 lines

```
const int K = 26;

struct Vertex {
    int next[K];
    bool output = false;
    int p = -1;
    char pch;
    int link = -1;
    int go[K];

    Vertex(int p=-1, char ch='$') : p(p), pch(ch) {
        fill(begin(next), end(next), -1);
        fill(begin(go), end(go), -1);
    }
};

vector<Vertex> t(1);

void add_string(string const& s) {
    int v = 0;
    for (char ch : s) {
        int c = ch - 'a';
        if (t[v].next[c] == -1) {
            t[v].next[c] = t.size();
            t.emplace_back(v, ch);
        }
        v = t[v].next[c];
    }
    t[v].output = true;
}

int go(int v, char ch);

int get_link(int v) {
    if (t[v].link == -1) {
        if (v == 0 || t[v].p == 0)
            t[v].link = 0;
        else
            t[v].link = go(get_link(t[v].p), t[v].pch);
    }
    return t[v].link;
}

int go(int v, char ch) {
    int c = ch - 'a';
    if (t[v].go[c] == -1) {
        if (t[v].next[c] != -1)
            t[v].go[c] = t[v].next[c];
        else
            t[v].go[c] = v == 0 ? 0 : go(get_link(v), ch);
    }
    return t[v].go[c];
}
```

## kmp.h
**Description:** TODO

4dfee5, 37 lines

```
int b[N];
int cnt = 0;

void knp_proc(string t,string p){
    int i=0,j=-1;b[0] = -1;
    while(i<(int)p.length()){
        while(j>=0 && p[i]!=p[j]) j =b[j];
        ++i;++j;
        b[i] = j;
    }
}

void knp_search(string t,string p){//count number of occurence
     of p in t
    int i=0,j=0;
    while(i<(int)t.length()){
        while(j>=0 && t[i] != p[j]) j = b[j];
        ++i;++j;
        if(j==(int)p.length()){
            ++cnt;
            j = b[j];
        }
    }
}

vector<int> prefix_function(string s) {
    int n = (int)s.length();
    vector<int> pi(n);
    for (int i = 1; i < n; i++) {
        int j = pi[i-1];
        while (j > 0 && s[i] != s[j])
            j = pi[j-1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}
```

## z-function.h
**Description:** TODO

9a2512, 18 lines

```
vector<int> z_function(string s) {
    int n = s.size();
    vector<int> z(n);
    int l = 0, r = 0;
    for(int i = 1; i < n; i++) {
        if(i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while(i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if(i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}
```

## manacher.h
**Description:** TODO

3069fe, 38 lines

```
//sub-palindrome queries
vector<int> manacher_odd(string s) {
    int n = s.size();
    s = "$" + s + "^";
    vector<int> p(n + 2);
    int l = 1, r = 1;
    for(int i = 1; i <= n; i++) {
        p[i] = max(0, min(r - i, p[l + (r - i)]));
        while(s[i - p[i]] == s[i + p[i]]) {
            p[i]++;
        }
        if(i + p[i] > r) {
            l = i - p[i], r = i + p[i];
        }
    }
    return vector<int>(begin(p) + 1, end(p) - 1);
}

vector<int> manacher(string s) {
    string t;
    for(auto c: s) {
        t += string("#") + c;
    }
    auto res = manacher_odd(t + "#");
    return vector<int>(begin(res) + 1, end(res) - 1);
}

//beware the range is inclusive and therefore not half-open -> [l, r]
bool is_palindrome(int l, int r, vector<int> &v){
    if(v[l +r] < (r - l + 1))return false;
    return true;
}

vector<int> build_manacher(string s){ //0 base index string
    auto v = manacher(s);
    for(auto&x:v)--x;
    return v;
}
```

## string-hashing.h
**Description:** TODO

ba27df, 17 lines

```
typedef __int128 HASH;
HASH p[NN], h[NN];
constexpr HASH M = 1000000000000000003;
mt19937 rng(chrono::steady_clock::now().time_since_epoch().
    count());//delete this if using my template
#define uid(a, b) uniform_int_distribution<long long>(a, b)(rng
    )//delete this if using my template

void compute_hash(string const& s){
    p[0] = 1, p[1] = uid(256, M-1);
    for(ll i=0;i<(int)s.length();++i){
        p[i+1] = p[i]*p[1]%M;
        h[i+1] = (h[i]*p[1] + s[i])%M;
    }
}

HASH sub_hash(ll l, ll r){ //range half-open [l, r)
    return (h[r] - p[r-l]*h[l]%M + M)%M;
}
```

## double-hashing-string.h
**Description:** TODO

276f01, 26 lines

```
//typedef
typedef long long ll;
typedef pair<ll,ll> pl;
#define M 1000000321
#define OP(x, y) pl operator x (pl a, pl b){return {a.first x b
    .first, (a.second y b.second) % M }; }
OP(+, +) OP(*, *) OP(-, + M -)
//random number generator
mt19937 gen(chrono::steady_clock::now().time_since_epoch().
    count());
uniform_int_distribution<ll> dist(256, M - 1);
//queries - check if S[i:i+l] == S[j:j+l](inclusive), S is a
    string, [:] is slice
#define H(i, l) (h[(i) + (l)] - h[i] * p[l])
#define EQ(i, j, l) (H(i, l) == H(j, l))
//preprocessing
const int N = 2e5;
string s;
pl p[N], h[N];
ll n;

int main(){
    cin >> n >> s;
    p[0] = {1,1}, p[1] = {dist(gen) | 1, dist(gen) };
    for(ll i=1;i<=(ll)s.length();++i){
        p[i] = p[i-1] * p[1];
        h[i] = h[i-1] * p[1] + make_pair(s[i-1], s[i-1]);
    }
}
```

## suffix-array.h
**Description:** TODO

4ca004, 53 lines

```
vector<int> sort_cyclic_shifts(string const& s) {
    int n = s.size();
    const int alphabet = 256;

    vector<int> p(n), c(n), cnt(max(alphabet, n), 0);
    for (int i = 0; i < n; i++)
        cnt[s[i]]++;
    for (int i = 1; i < alphabet; i++)
        cnt[i] += cnt[i-1];
    for (int i = 0; i < n; i++)
        p[--cnt[s[i]]] = i;
    c[p[0]] = 0;
    int classes = 1;
    for (int i = 1; i < n; i++) {
        if (s[p[i]] != s[p[i-1]])
            classes++;
        c[p[i]] = classes - 1;
    }

    vector<int> pn(n), cn(n);
    for (int h = 0; (1 << h) < n; ++h) {
        for (int i = 0; i < n; i++) {
            pn[i] = p[i] - (1 << h);
            if (pn[i] < 0)
                pn[i] += n;
        }
        fill(cnt.begin(), cnt.begin() + classes, 0);
        for (int i = 0; i < n; i++)
            cnt[c[pn[i]]]++;
        for (int i = 1; i < classes; i++)
            cnt[i] += cnt[i-1];
        for (int i = n-1; i >= 0; i--)
            p[--cnt[c[pn[i]]]] = pn[i];
        cn[p[0]] = 0;
        classes = 1;
        for (int i = 1; i < n; i++) {
            pair<int, int> cur = {c[p[i]], c[(p[i] + (1 << h))
                % n]};
            pair<int, int> prev = {c[p[i-1]], c[(p[i-1] + (1 <<
                h)) % n]};
            if (cur != prev)
                ++classes;
```

```
            cn[p[i]] = classes - 1;
        }
        c.swap(cn);
    }
    return p;
}

vector<int> suffix_array_construction(string s) {
    s += "$";
    vector<int> sorted_shifts = sort_cyclic_shifts(s);
    sorted_shifts.erase(sorted_shifts.begin());
    return sorted_shifts;
}
```

### suffix-automaton.h

**Description:** TODO

fe30a6, 52 lines

```
struct state {
    int len, link;
    map<char, int> next;
    map<char, int> go;
};

const int MAXLEN = 100001; //this one is 10^5 becareful
state st[MAXLEN * 2];
int sz, last;

void sa_init() {
    st[0].len = 0;
    st[0].link = -1;
    sz++;
    last = 0;
}

void sa_extend(char c) {
    int cur = sz++;
    st[cur].len = st[last].len + 1;
    int p = last;
    while (p != -1 && !st[p].next.count(c)) {
        st[p].next[c] = cur;
        p = st[p].link;
    }
    if (p == -1) {
        st[cur].link = 0;
    } else {
        int q = st[p].next[c];
        if (st[p].len + 1 == st[q].len) {
            st[cur].link = q;
        } else {
            int clone = sz++;
            st[clone].len = st[p].len + 1;
            st[clone].next = st[q].next;
            st[clone].link = st[q].link;
            while (p != -1 && st[p].next[c] == q) {
                st[p].next[c] = clone;
                p = st[p].link;
            }
            st[q].link = st[cur].link = clone;
        }
    }
    last = cur;
}

int sa_go(int v, char c) {
    if (st[v].go.count(c)) return st[v].go[c];
    if (st[v].next.count(c)) return st[v].go[c] = st[v].next[c
        ];
    if (st[v].link == -1) return st[v].go[c] = 0; // fallback
        to root
    return st[v].go[c] = sa_go(st[v].link, c);
}
```

### trie.h

**Description:** TODO

e121b2, 26 lines

```
// string trie structure
const int K = 26;

struct node{
    int to[K];
    bool output = false;

    node(){
        fill(begin(to), end(to), -1);
    }
};

vector<node>trie(1);

void add(const& string s){
    int v = 0;
    for(char ch: s){
        int c = ch - 'a';
        if(trie[v][c] == -1){
            trie[v][c] = trie.size();
            trie.push_back(node());
        }
        v = trie[v][c];
    }
    trie[v].output = true;
}
```

### lyndon-factorization.h

**Description:** TODO

a3cc8c, 40 lines

```
vector<string> duval(string const& s) {
    int n = s.size();
    int i = 0;
    vector<string> factorization;
    while (i < n) {
        int j = i + 1, k = i;
        while (j < n && s[k] <= s[j]) {
            if (s[k] < s[j])
                k = i;
            else
                k++;
            j++;
        }
        while (i <= k) {
            factorization.push_back(s.substr(i, j - k));
            i += j - k;
        }
    }
    return factorization;
}

string min_cyclic_string(string s) {
    s += s;
    int n = s.size();
    int i = 0, ans = 0;
    while (i < n / 2) {
        ans = i;
        int j = i + 1, k = i;
        while (j < n && s[k] <= s[j]) {
            if (s[k] < s[j])
                k = i;
            else
                k++;
            j++;
        }
        while (i <= k)
            i += j - k;
    }
    return s.substr(ans, n / 2);
}
```

## 2.9 Computational Geometry

### geometry.h

**Description:** TODO

963fd3, 167 lines

```
//Geometry
#pragma GCC target("avx2")
#pragma GCC optimize("O3")
#pragma GCC optimize("unroll-loops")

typedef long long int ll;
typedef long double ld;
typedef complex<ld> pt;
struct line {
  pt P, D; bool S = false;
  line(pt p, pt q, bool b = false) : P(p), D(q - p), S(b) {}
  line(pt p, ld th) : P(p), D(polar((ld)1, th)) {}
};
struct circ { pt C; ld R; };

#define X real()
#define Y imag()
#define CRS(a, b) (conj(a) * (b)).Y //scalar cross product
#define DOT(a, b) (conj(a) * (b)).X //dot product
#define U(p) ((p) / abs(p)) //unit vector in direction of p (
    don't use if Z(p) == true)
#define Z(x) (abs(x) < EPS)
#define A(a) (a).begin(), (a).end() //shortens sort(),
    upper_bound(), etc. for vectors

//constants (INF and EPS may need to be modified)
ld PI = acosl(-1), INF = 1e20, EPS = 1e-12;
pt I = {0, 1};

//true if d1 and d2 parallel (zero vectors considered parallel
    to everything)
bool parallel(pt d1, pt d2) { return Z(d1) || Z(d2) || Z(CRS(U(
    d1), U(d2))); }

//"above" here means if l & p are rotated such that l.D points
    in the +x direction, then p is above l. Returns arbitrary
    boolean if p is on l
bool above_line(pt p, line l) { return CRS(p - l.P, l.D) > 0; }

//true if p is on line l
bool on_line(pt p, line l) { return parallel(l.P - p, l.D) &&
    (!l.S || DOT(l.P - p, l.P + l.D - p) <= EPS); }

//returns 0 for no intersection, 2 for infinite intersections,
    1 otherwise. p holds intersection pt
ll intsct(line l1, line l2, pt& p) {
  if(parallel(l1.D, l2.D)) //note that two parallel segments
      sharing one endpoint are considered to have infinite
      intersections here
    return 2 * (on_line(l1.P, l2) || on_line(l1.P + l1.D, l2)
        || on_line(l2.P, l1) || on_line(l2.P + l2.D, l1));
  pt q = l1.P + l1.D * CRS(l2.D, l2.P - l1.P) / CRS(l2.D, l1.D)
      ;
  if(on_line(q, l1) && on_line(q, l2)) { p = q; return 1; }
  return 0;
}
```

```
//closest pt on l to p
pt cl_pt_on_l(pt p, line l) {
  pt q = l.P + DOT(U(l.D), p - l.P) * U(l.D);
  if(on_line(q, l)) return q;
  return abs(p - l.P) < abs(p - l.P - l.D) ? l.P : l.P + l.D;
}

//distance from p to l
ld dist_to(pt p, line l) { return abs(p - cl_pt_on_l(p, l)); }

//p reflected over l
pt refl_pt(pt p, line l) { return conj((p - l.P) / U(l.D)) * U(
    l.D) + l.P; }

//ray r reflected off l (if no intersection, returns original
    ray)
line reflect_line(line r, line l) {
  pt p; if(intsct(r, l, p) - 1) return r;
  return line(p, p + INF * (p - refl_pt(r.P, l)), 1);
}

//altitude from p to l
line alt(pt p, line l) { l.S = 0; return line(p, cl_pt_on_l(p,
    l)); }

//angle bisector of angle abc
line ang_bis(pt a, pt b, pt c) { return line(b, b + INF * (U(a
    - b) + U(c - b)), 1); }

//perpendicular bisector of l (assumes l.S == 1)
line perp_bis(line l) { return line(l.P + l.D / (ld)2, arg(l.D
    * I)); }

//orthocenter of triangle abc
pt orthocent(pt a, pt b, pt c) { pt p; intsct(alt(a, line(b, c)
    ), alt(b, line(a, c)), p); return p; }

//incircle of triangle abc
circ incirc(pt a, pt b, pt c) {
  pt cent; intsct(ang_bis(a, b, c), ang_bis(b, a, c), cent);
  return {cent, dist_to(cent, line(a, b))};
}

//circumcircle of triangle abc
circ circumcirc(pt a, pt b, pt c) {
  pt cent; intsct(perp_bis(line(a, b, 1)), perp_bis(line(a, c,
      1)), cent);
  return {cent, abs(cent - a)};
}

//is pt p inside the (not necessarily convex) polygon given by
    poly
bool in_poly(pt p, vector<pt>& poly) {
  line l = line(p, {INF, INF * PI}, 1);
  bool ans = false;
  pt lst = poly.back(), tmp;
  for(pt q : poly) {
    line s = line(q, lst, 1); lst = q;
    if(on_line(p, s)) return false; //change if border included
    else if(intsct(l, s, tmp)) ans = !ans;
  }
  return ans;
}

//area of polygon, vertices in order (cw or ccw)
ld area(vector<pt>& poly) {
  ld ans = 0;
  pt lst = poly.back();
  for(pt p : poly) ans += CRS(lst, p), lst = p;
  return abs(ans / 2);
}

//perimeter of polygon, vertices in order (cw or ccw)
ld perim(vector<pt>& poly) {
  ld ans = 0;
  pt lst = poly.back();
  for(pt p : poly) ans += abs(lst - p), lst = p;
  return ans;
}

//centroid of polygon, vertices in order (cw or ccw)
pt centroid(vector<pt>& poly) {
  ld area = 0;
  pt lst = poly.back(), ans = {0, 0};
  for(pt p : poly) {
    area += CRS(lst, p);
    ans += CRS(lst, p) * (lst + p) / (ld)3;
    lst = p;
  }
  return ans / area;
}

//invert a point over a circle (doesn't work for center of
    circle)
pt circInv(pt p, circ c) {
    return c.R * c.R / conj(p - c.C) + c.C;
}

//vector of intersection pts of two circs (up to 2) (if circles
    same, returns empty vector)
vector<pt> intsctCC(circ c1, circ c2) {
  if(c1.R < c2.R) swap(c1, c2);
  pt d = c2.C - c1.C;
  if(Z(abs(d) - c1.R - c2.R)) return {c1.C + polar(c1.R, arg(c2
      .C - c1.C))};
  if(!Z(d) && Z(abs(d) - c1.R + c2.R)) return {c1.C + c1.R * U(
      d)};
  if(abs(abs(d) - c1.R) >= c2.R - EPS) return {};
  ld th = acosl((c1.R * c1.R + norm(d) - c2.R * c2.R) / (2 * c1
      .R * abs(d)));
  return {c1.C + polar(c1.R, arg(d) + th), c1.C + polar(c1.R,
      arg(d) - th)};
}

//vector of intersection pts of a line and a circ (up to 2)
vector<pt> intsctCL(circ c, line l) {
  vector<pt> v, ans;
  if(parallel(l.D, c.C - l.P)) v = {c.C + c.R * U(l.D), c.C - c
      .R * U(l.D)};
  else v = intsctCC(c, circ{refl_pt(c.C, l), c.R});
  for(pt p : v) if(on_line(p, l)) ans.push_back(p);
  return ans;
}

//external tangents of two circles (negate c2.R for internal
    tangents)
vector<line> circTangents(circ c1, circ c2) {
  pt d = c2.C - c1.C;
  ld dr = c1.R - c2.R, d2 = norm(d), h2 = d2 - dr * dr;
  if(Z(d2) || h2 < 0) return {};
  vector<line> ans;
  for(ld sg : {-1, 1}) {
    pt u = (d * dr + d * I * sqrt(h2) * sg) / d2;
    ans.push_back(line(c1.C + u * c1.R, c2.C + u * c2.R, 1));
  }
  if(Z(h2)) ans.pop_back();
  return ans;
}
```

closest-pair-of-points.h

**Description:** TODO

85eeeb, 67 lines

```
struct pt {
    int x, y, id;
};

struct cmp_x {
    bool operator()(const pt & a, const pt & b) const {
        return a.x < b.x || (a.x == b.x && a.y < b.y);
    }
};

struct cmp_y {
    bool operator()(const pt & a, const pt & b) const {
        return a.y < b.y;
    }
};

int n;
vector<pt> a;

double mindist;
pair<int, int> best_pair;

void upd_ans(const pt & a, const pt & b) {
    double dist = sqrt((a.x - b.x)*(a.x - b.x) + (a.y - b.y)*(a
        .y - b.y));
    if (dist < mindist) {
        mindist = dist;
        best_pair = {a.id, b.id};
    }
}

vector<pt> t;

void rec(int l, int r) {
    if (r - l <= 3) {
        for (int i = l; i < r; ++i) {
            for (int j = i + 1; j < r; ++j) {
                upd_ans(a[i], a[j]);
            }
        }
        sort(a.begin() + l, a.begin() + r, cmp_y());
        return;
    }

    int m = (l + r) >> 1;
    int midx = a[m].x;
    rec(l, m);
    rec(m, r);

    merge(a.begin() + l, a.begin() + m, a.begin() + m, a.begin
        () + r, t.begin(), cmp_y());
    copy(t.begin(), t.begin() + r - l, a.begin() + l);

    int tsz = 0;
    for (int i = l; i < r; ++i) {
        if (abs(a[i].x - midx) < mindist) {
            for (int j = tsz - 1; j >= 0 && a[i].y - t[j].y <
                mindist; --j)
                upd_ans(a[i], t[j]);
            t[tsz++] = a[i];
        }
    }
}

void call_rec(){
    t.resize(n);
```

```
    sort(a.begin(), a.end(), cmp_x());
    mindist = 1E20;
    rec(0, n);
}
```

## 2.10 Advanced Algorithms

### fft.h

**Description:** TODO

cbf4b6, 63 lines

```
//from cp algorithm the inplace-computation fft part
using cd = complex<double>;
const double PI = acos(-1);

int reverse(int num, int lg_n) {
    int res = 0;
    for (int i = 0; i < lg_n; i++) {
        if (num & (1 << i))
            res |= 1 << (lg_n - 1 - i);
    }
    return res;
}

void fft(vector<cd> & a, bool invert) {
    int n = a.size();
    int lg_n = 0;
    while ((1 << lg_n) < n)
        lg_n++;

    for (int i = 0; i < n; i++) {
        if (i < reverse(i, lg_n))
            swap(a[i], a[reverse(i, lg_n)]);
    }

    for (int len = 2; len <= n; len <<= 1) {
        double ang = 2 * PI / len * (invert ? -1 : 1);
        cd wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd w(1);
            for (int j = 0; j < len / 2; j++) {
                cd u = a[i+j], v = a[i+j+len/2] * w;
                a[i+j] = u + v;
                a[i+j+len/2] = u - v;
                w *= wlen;
            }
        }
    }

    if (invert) {
        for (cd & x : a)
            x /= n;
    }
}

vector<int> multiply(vector<int> const& a, vector<int> const& b
    ) {
    vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1;
    while (n < (int)a.size() + (int)b.size())
        n <<= 1;
    fa.resize(n);
    fb.resize(n);

    fft(fa, false);
    fft(fb, false);
    for (int i = 0; i < n; i++)
        fa[i] *= fb[i];
    fft(fa, true);

    vector<int> result(n);
    for (int i = 0; i < n; i++)
        result[i] = round(fa[i].real());
    return result;
}
```

### mo2D.h

**Description:** TODO

9df2fb, 49 lines

```
/* always use zero based indexing*/
/* there N/B blocks so left move ~ N/B*B + Q*B, right move ~ N/
    B*B + (N/B)*B */
/* so O(N^2/B + N + QB) ~~ O((N+Q)sqrt(N)) when B = sqrt(N) */
/* when Q~N it's ~ O(N^3/2) */
void remove(int idx);  // TODO: remove value at idx from data
    structure
void add(int idx);     // TODO: add value at idx from data
    structure
int get_answer();  // TODO: extract the current answer of the
    data structure

int block_size;

struct Query {
    int l, r, idx;
    bool operator<(Query other) const
    {
        return make_pair(l / block_size, r) <
               make_pair(other.l / block_size, other.r);
    }
};

vector<int> mo_s_algorithm(vector<Query>& queries) {
    vector<int> answers(queries.size());
    sort(queries.begin(), queries.end());

    // TODO: initialize data structure

    int cur_l = 0;
    int cur_r = -1;
    // invariant: data structure will always reflect the range
        [cur_l, cur_r]
    for (Query q : queries) {
        while (cur_l > q.l) {
            cur_l--;
            add(cur_l);
        }
        while (cur_r < q.r) {
            cur_r++;
            add(cur_r);
        }
        while (cur_l < q.l) {
            remove(cur_l);
            cur_l++;
        }
        while (cur_r > q.r) {
            remove(cur_r);
            cur_r--;
        }
        answers[q.idx] = get_answer();
    }
    return answers;
}
```

### mo3D.h

**Description:** TODO

dc4a5e, 62 lines

```
/* always use zero based indexing*/
/* ~ O(N^(5/3)) */
int block_size;

struct Query {
    int l, r, t, idx;
    bool operator<(Query other) const
    {
        return make_tuple(l / block_size, r / block_size, t) <
               make_tuple(other.l / block_size, other.r /
                    block_size,other.t);
    }
};

int ans[N];
void mo_s_algorithm(vector<Query>& queries) {
    sort(queries.begin(), queries.end());
    // TODO: initialize data structure

    int cur_l = 0;
    int cur_r = -1;
    int cur_t = -1;
    // invariant: data structure will always reflect the range
        [cur_l, cur_r]
    for (Query q : queries) {
        while (cur_l > q.l) {
            cur_l--;
            //add(cur_l)
        }
        while (cur_r < q.r) {
            cur_r++;
            //add(cur_r)
        }
        while (cur_l < q.l) {
            //remove(cur_l);
            cur_l++;
        }
        while (cur_r > q.r) {
            //remove(cur_r);
            cur_r--;
        }

        while(cur_t < q.t){
            cur_t++;
            //update
        }
        while(cur_t > q.t){
            //update rollback
            cur_t--;
        }
        ans[q.idx] = get_answer(); // declare array for this
    }
}

void getBlockSize(int n, int q){
    if (q >= n) {
        // most common: Q ~ N
        block_size = max(1, (int)pow(n, 2.0/3.0));
    } else {
        // general cube-root rule
        block_size = max(1, (int)pow((long double)n*n / q,
            1.0/3.0));
    }
    // then sort your queries by (L/B, R/B, T) or via Hilbert
}
```

### divide-and-conquer-dp.h

**Description:** TODO

108b88, 55 lines

```
//TODO: initialize dp , initialize cost() function of use slide
    ()
//ll dp[N][M];
```

```
//generic implementation for sliding range technique for logn
     cost()
//(persistent segtree alternative)
ll ccost = 0;
int cl = 0, cr = -1;
void slide(int l, int r){
    while(cr < r){
        ++cr;
        //add();
        //...
    }
    while(cl > l){
        --cl;
        //add();
        //...
    }
    while(cr > r){
        //remove();
        //...
        --cr;
    }
    while(cl < l){
        //remove();
        //...
        ++cl;
    }
}

void compute(int l, int r, int optl, int optr, int j){
    if(l>r)return;

    int mid = (l+r)>>1;
    //pair<ll, int> best = {0,-1};
    //pair<ll, int> best = {LINF,-1};

    //dp is satisfy quadrangle IE if cost() satisfy qudrangle
         IE
    //if cost() is QF => opt() is nondecreasing
    for(int k=optl;k<=min(mid,optr);++k){
        slide(k,mid);
        //best = max(best, {((k>0)?dp[k-1][j-1]:0) + ccost, k }
             );
        //best = min(best, {((k>0)?dp[k-1][j-1]:0) + ccost, k }
             );
    }

    //dp[mid][j] = max(dp[mid][j], best.first);
    //dp[mid][j] = min(dp[mid][j], best.first);
    int opt = best.second;
    if(l!=r){
        compute(l,mid-1,optl,opt,j);
        compute(mid+1,r,opt,optr,j);
    }
}

//TODO: set dp to LINF or -LINF
```

# cellul4r (3)

## 2SAT.h

**Description:** solve 2 sat form of or to implies
**Time:** $\mathcal{O}(N+M)$

025007, 70 lines

```
struct TwoSatSolver {
    int n_vars;
    int n_vertices;
    vector<vector<int>> adj, adj_t;
    vector<bool> used;
    vector<int> order, comp;
    vector<bool> assignment;
    TwoSatSolver(int _n_vars) : n_vars(_n_vars), n_vertices(2 *
         n_vars), adj(n_vertices), adj_t(n_vertices), used(
        n_vertices), order(), comp(n_vertices, -1), assignment
        (n_vars) {
        order.reserve(n_vertices);
    }
    void dfs1(int v) {
        used[v] = true;
        for (int u : adj[v]) {
            if (!used[u])
                dfs1(u);
        }
        order.push_back(v);
    }
    void dfs2(int v, int cl) {
        comp[v] = cl;
        for (int u : adj_t[v]) {
            if (comp[u] == -1)
                dfs2(u, cl);
        }
    }
    bool solve_2SAT() {
        order.clear();
        used.assign(n_vertices, false);
        for (int i = 0; i < n_vertices; ++i) {
            if (!used[i])
                dfs1(i);
        }
        comp.assign(n_vertices, -1);
        for (int i = 0, j = 0; i < n_vertices; ++i) {
            int v = order[n_vertices - i - 1];
            if (comp[v] == -1)
                dfs2(v, j++);
        }

        assignment.assign(n_vars, false);
        for (int i = 0; i < n_vertices; i += 2) {
            if (comp[i] == comp[i + 1])
                return false;
            assignment[i / 2] = comp[i] > comp[i + 1];
        }
        return true;
    }
    void add_disjunction(int a, bool na, int b, bool nb) {
        // na and nb signify whether a and b are to be negated
        a = 2 * a ^ na;
        b = 2 * b ^ nb;
        int neg_a = a ^ 1;
        int neg_b = b ^ 1;
        adj[neg_a].push_back(b);
        adj[neg_b].push_back(a);
        adj_t[b].push_back(neg_a);
        adj_t[a].push_back(neg_b);
    }
    static void example_usage() {
        TwoSatSolver solver(3);
        solver.add_disjunction(0, false, 1, true);
        //     a  or  not b
        solver.add_disjunction(0, true, 1, true);   // not a
             or  not b
        solver.add_disjunction(1, false, 2, false); //     b
             or      c
        solver.add_disjunction(0, false, 0, false); //     a
             or      a
        assert(solver.solve_2SAT() == true);
        auto expected = vector<bool>(True, False, True);
        assert(solver.assignment == expected);
    }
};
```

## kadane2D.h

**Time:** $\mathcal{O}()$

6227c1, 59 lines

```
int kadane(vector<int>& temp) {
    int rows = temp.size();
    int currSum = 0;
    int maxSum = INT_MIN;

    for (int i = 0; i < rows; i++) {
        currSum += temp[i];

        // Update maxSum if the current sum is greater
        if (maxSum < currSum) {
            maxSum = currSum;
        }

        // If the current sum becomes negative, reset it to 0
        if (currSum < 0) {
            currSum = 0;
        }
    }

    return maxSum;
}

// Function to find the maximum sum rectangle in a 2D matrix
int maxRectSum(vector<vector<int>> &mat) {
    int rows = mat.size();
    int cols = mat[0].size();

    int maxSum = INT_MIN;

    // Initialize a temporary array to store row-wise
    // sums between left and right boundaries
    vector<int> temp(rows);

    // Check for all possible left and right boundaries
    for (int left = 0; left < cols; left++) {

        // Reset the temporary array for each new left boundary
        for (int i = 0; i < rows; i++)
            temp[i] = 0;

        for (int right = left; right < cols; right++) {

            // Update the temporary array with the current
            // column's values
            for (int row = 0; row < rows; row++) {
                temp[row] += mat[row][right];
            }

            // Find the maximum sum of the subarray for the
            // current column boundaries
            int sum = kadane(temp);

            // Update the maximum sum found so far
            maxSum = max(maxSum, sum);
        }
    }

    return maxSum;
}
```

LCS.h
**Time:** $\mathcal{O}()$
7b1c80, 90 lines

```
int lcs(string &s1, string &s2) {
    int m = s1.size();
    int n = s2.size();

    // Initializing a matrix of size (m+1)*(n+1)
    vector<vector<int>> dp(m + 1, vector<int>(n + 1, 0));

    // Building dp[m+1][n+1] in bottom-up fashion
    for (int i = 1; i <= m; ++i) {
        for (int j = 1; j <= n; ++j) {
            if (s1[i - 1] == s2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
        }
    }

    // dp[m][n] contains length of LCS for s1[0..m-1]
    // and s2[0..n-1]
    return dp[m][n];
}

int lcs(string &s1, string &s2) {

    int m = s1.length(), n = s2.length();

    // dp vector is initialized to all zeros
    // This vector stores the LCS values for the current row.
    // dp[j] represents LCS of s1[0..i] and s2[0..j]
    vector<int> dp(n + 1, 0);

    // i and j represent the lengths of s1 and s2 respectively
    for (int i = 1; i <= m; ++i) {

        // prev stores the value from the previous
        // row and previous column (i-1), (j -1)
        // Used to keep track of LCS[i-1][j-1] while updating
             dp[j]
        int prev = dp[0];

        for (int j = 1; j <= n; ++j) {

            // temp temporarily stores the current
            // dp[j] before it gets updated
            int temp = dp[j];

            // If characters match, add 1 to the value
            // from the previous row and previous column
            // dp[j] = 1 + LCS[i-1][j-1]
            if (s1[i - 1] == s2[j - 1])
                dp[j] = 1 + prev;
            else
                // Otherwise, take the maximum of the
                // left (dp[j-1]) and top (dp[j]) values
                dp[j] = max(dp[j - 1], dp[j]);

            // Update prev for the next iteration
            // This keeps the value of the previous
          // row (i-1) for future comparisons
            prev = temp;
        }
    }

    // The last element of the vector contains the length of
         the LCS
    // dp[n] stores the length of LCS of s1[0..m] and s2[0..n]
    return dp[n];
}

// Function that returns the number of paths
int numberofPaths(int source, int destination, int n, int fre
     [])
{

// make topological sorting
    vector<int> s = topological_sorting(fre, n);

    // to store required answer.
    int dp[n] = { 0 };

    // answer from destination
    // to destination is 1.
    dp[destination] = 1;

    // traverse in reverse order
    for (int i = s.size() - 1; i >= 0; i--) {
        for (int j = 0; j < v[s[i]].size(); j++) {
            dp[s[i]] += dp[v[s[i]][j]];
        }
    }

    return dp[source];
}
```

# Mathematics (4)

## 4.1 Equations

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The extremum is given by $x = -b/2a$.

$$\begin{matrix} ax + by = e \\ cx + dy = f \end{matrix} \Rightarrow \begin{matrix} x = \dfrac{ed - bf}{ad - bc} \\ y = \dfrac{af - ec}{ad - bc} \end{matrix}$$

In general, given an equation $Ax = b$, the solution to a variable $x_i$ is given by

$$x_i = \frac{\det A_i'}{\det A}$$

where $A_i'$ is $A$ with the $i$'th column replaced by $b$.

Non-distinct roots $r$ become polynomial factors, e.g. $a_n = (d_1 n + d_2) r^n$.

## 4.2 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w$$
$$\cos(v + w) = \cos v \cos w - \sin v \sin w$$
$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$
$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2}$$
$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2}$$

$$(V + W) \tan(v - w)/2 = (V - W) \tan(v + w)/2$$

where $V, W$ are lengths of sides opposite angles $v, w$.

$$a \cos x + b \sin x = r \cos(x - \phi)$$
$$a \sin x + b \cos x = r \sin(x + \phi)$$

where $r = \sqrt{a^2 + b^2}, \phi = \text{atan2}(b, a)$.

## 4.3 Geometry

### 4.3.1 Triangles

Side lengths: $a, b, c$

Semiperimeter: $p = \dfrac{a + b + c}{2}$

Area: $A = \sqrt{p(p - a)(p - b)(p - c)}$

Circumradius: $R = \dfrac{abc}{4A}$

Inradius: $r = \dfrac{A}{p}$

Length of median (divides triangle into two equal-area triangles): $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Length of bisector (divides angles in two): $s_a = \sqrt{bc\left[1 - \left(\dfrac{a}{b + c}\right)^2\right]}$

Law of sines: $\dfrac{\sin \alpha}{a} = \dfrac{\sin \beta}{b} = \dfrac{\sin \gamma}{c} = \dfrac{1}{2R}$

Law of cosines: $a^2 = b^2 + c^2 - 2bc \cos \alpha$

Law of tangents: $\dfrac{a + b}{a - b} = \dfrac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}}$

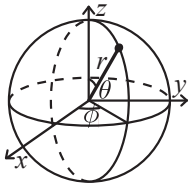
### 4.3.2 Quadrilaterals

With side lengths $a, b, c, d$, diagonals $e, f$, diagonals angle $\theta$, area $A$ and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2 f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is $180°$, $ef = ac + bd$, and $A = \sqrt{(p - a)(p - b)(p - c)(p - d)}$.

### 4.3.3 Spherical coordinates




$$\begin{array}{cc} x = r\sin\theta\cos\phi & r = \sqrt{x^2+y^2+z^2} \\ y = r\sin\theta\sin\phi & \theta = \operatorname{acos}(z/\sqrt{x^2+y^2+z^2}) \\ z = r\cos\theta & \phi = \operatorname{atan2}(y,x) \end{array}$$

## 4.4 Derivatives/Integrals

$$\begin{array}{cc} \frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}\tan x = 1+\tan^2 x & \frac{d}{dx}\arctan x = \frac{1}{1+x^2} \\ \int \tan ax = -\frac{\ln|\cos ax|}{a} & \int x\sin ax = \frac{\sin ax - ax\cos ax}{a^2} \\ \int e^{-x^2} = \frac{\sqrt{\pi}}{2}\operatorname{erf}(x) & \int xe^{ax}dx = \frac{e^{ax}}{a^2}(ax-1) \end{array}$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

## 4.5 Sums

$$c^a + c^{a+1} + \cdots + c^b = \frac{c^{b+1}-c^a}{c-1}, c \neq 1$$

$$\begin{aligned} 1+2+3+\cdots+n &= \frac{n(n+1)}{2} \\ 1^2+2^2+3^2+\cdots+n^2 &= \frac{n(2n+1)(n+1)}{6} \\ 1^3+2^3+3^3+\cdots+n^3 &= \frac{n^2(n+1)^2}{4} \\ 1^4+2^4+3^4+\cdots+n^4 &= \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \end{aligned}$$

## 4.6 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots, (-\infty < x < \infty)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots, (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \ldots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots, (-\infty < x < \infty)$$

## 4.7 Probability theory

Let $X$ be a discrete random variable with probability $p_X(x)$ of assuming the value $x$. It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x xp_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where $\sigma$ is the standard deviation. If $X$ is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent $X$ and $Y$,

$$V(aX + bY) = a^2V(X) + b^2V(Y).$$

### 4.7.1 Discrete distributions

#### Binomial distribution

The number of successes in $n$ independent yes/no experiments, each which yields success with probability $p$ is $\operatorname{Bin}(n,p)$, $n = 1, 2, \ldots$, $0 \leq p \leq 1$.

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu = np,\ \sigma^2 = np(1-p)$$

$\operatorname{Bin}(n,p)$ is approximately $\operatorname{Po}(np)$ for small $p$.

#### First success distribution

The number of trials needed to get the first success in independent yes/no experiments, each which yields success with probability $p$ is $\operatorname{Fs}(p)$, $0 \leq p \leq 1$.

$$p(k) = p(1-p)^{k-1},\ k = 1, 2, \ldots$$

$$\mu = \frac{1}{p},\ \sigma^2 = \frac{1-p}{p^2}$$

#### Poisson distribution

The number of events occurring in a fixed period of time $t$ if these events occur with a known average rate $\kappa$ and independently of the time since the last event is $\operatorname{Po}(\lambda)$, $\lambda = t\kappa$.

$$p(k) = e^{-\lambda}\frac{\lambda^k}{k!}, k = 0, 1, 2, \ldots$$

$$\mu = \lambda,\ \sigma^2 = \lambda$$

### 4.7.2 Continuous distributions

#### Uniform distribution

If the probability density function is constant between $a$ and $b$ and 0 elsewhere it is $\operatorname{U}(a,b)$, $a < b$.

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = \frac{a+b}{2},\ \sigma^2 = \frac{(b-a)^2}{12}$$

#### Exponential distribution

The time between events in a Poisson process is $\operatorname{Exp}(\lambda)$, $\lambda > 0$.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\mu = \frac{1}{\lambda},\ \sigma^2 = \frac{1}{\lambda^2}$$

#### Normal distribution

Most real random values with mean $\mu$ and variance $\sigma^2$ are well described by $\mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ then

$$aX_1 + bX_2 + c \sim \mathcal{N}(\mu_1 + \mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$

# Number Theoretic Functions

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## Contents





## 1 Multiplicative Function

**Definition 1.1** (Multiplicative Function)**.** *A number-theoretic function $f$ is **multiplicative** if $f(1) = 1$ and $f(mn) = f(m)f(n)$, $\forall m, n \in \mathbb{N}$ such that $gcd(m, n) = 1$. Additionally, $f$ is called **completely multiplicative** if $f(mn) = f(m)f(n)$, $\forall m, n \in \mathbb{N}$.*

**Examples.**

- $1(n) = 1$: The constant function.
- $Id(n) = n$: The identity function.
- $\varepsilon(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$: The unit function
- $\tau(n) = \sum_{d|n} 1$: The number of divisors function.
- $\sigma(n) = \sum_{d|n} d$: The sum of divisors function.
- $\phi(n) = \sum_{i=1}^{n} [gcd(i, n) = 1]$: Euler's Totient Function (here the third brackets serve as a boolean function, which returns 1 if the condition is true, 0 otherwise.)

Note that $1, Id, \varepsilon$ are completely multiplicative as well, while $\phi, \tau, \sigma$ aren't.

**Theorem 1.** *If $f$ is a multiplicative function and if $n = \prod_{i=1}^{r} p_i^{e_i}$, then $f(n) = \prod_{i=1}^{r} f(p_i^{e_i})$.*

*Proof.* Since $gcd(p_i^{e_i}, p_j^{e_j}) = 1$, $\forall i \neq j$, induction on $r$ proves the theorem. □



## 2 Dirichlet Convolution

**Definition 2.1** (Dirichlet Convolution)**.** *If $f$ and $g$ are two arithmetic functions, then their **dirichlet convolution**, denoted by $f * g$, is defined as*

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d)$$

*Alternatively, we can write $(f * g)(n) = \sum_{ab=n} f(a)g(b)$.*

### Properties of Dirichlet Convolution

- Convolution is Commutative: $f * g = g * f$
- It is Associative: $(f * g) * h = f * (g * h)$

  *Proof.* $((f * g) * h)(n) = \sum_{dc=n} (f * g)(d)h(c) = \sum_{dc=n} \sum_{ab=d} f(a)g(b)h(c) = \sum_{abc=n} f(a)g(b)h(c)$

  Similarly, $(f * (g * h))(n) = \sum_{ad=n} f(a)(g * h)(d) = \sum_{ad=n} f(a) \sum_{bc=d} g(b)h(c) = \sum_{abc=n} f(a)g(b)h(c)$ □

- $f * \varepsilon = f$

  *Proof.*
  $$(f * \varepsilon)(n) = \sum_{d|n} f(d)\varepsilon(n/d)$$
  Now if $d < n$, $n/d > 1$, so $\varepsilon(n/d) = 0$. Therefore
  $$(f * \varepsilon)(n) = f(n)\varepsilon(1) = f(n)$$
  □

**Theorem 2.** *if $f$ and $g$ are both multiplicative, so is $f * g$.*

*Proof.* Let $h = f * g$. Now, if $gcd(m, n) = 1$,

$$h(mn) = \sum_{d|mn} f(d)g(mn/d)$$

Now since $d|mn$ and $gcd(m, n) = 1$, $d = ab$ where $a|m, b|n$ and $gcd(a, b) = 1$. Hence,

$$\begin{aligned} h(mn) &= \sum_{a|m,b|n} f(ab)g(mn/ab) \\ &= \sum_{a|m,b|n} f(a)f(b)g(m/a)g(n/b) \\ &= \sum_{a|m} f(a)g(m/a) \sum_{b|n} f(b)g(n/b) \\ &= h(m)h(n) \end{aligned}$$

□

**Corollary 1.** *if $f$ is a multiplicative function, and $F(n) = \sum_{d|n} f(n)$, $F$ is multiplicative as well.*

*Proof.* $F(n) = \sum_{d|n} f(n) = \sum_{d|n} f(n)1(\frac{n}{d}) = (f * 1)(n)$.
Since $f, 1$ both are multiplicative, by **theorem 2**, $F$ is also multiplicative. □

**Theorem 3.** *if $h = f * g$ and $h, g$ are both multiplicative, so is $f$.*

*Proof.* Suppose $f$ is not multiplicative. So, there exists a pair of positive integers (m, n) with $gcd(m, n) = 1$ such that $f(mn) \neq f(m)f(n)$. We take such a pair with smallest $mn$.

If $mn = 1$ then $f(1) \neq f(1)f(1)$ which implies $f(1) \neq 1$. Now since $g$ is multiplicative, $g(1) = 1$. But $h(1) = (f*g)(1) = f(1)g(1) \neq 1$ since $f(1) \neq 1$. This is a contradiction since $h$ is multiplicative and $h(1) = 1$.



If $mn > 1$ then

$$\begin{aligned} h(mn) &= \sum_{d|mn} f(d)g(mn/d) \\ &= \sum_{a|m,b|n} f(ab)g(mn/ab) \\ &= \sum_{a|m,b|n,ab<mn} f(a)f(b)g(m/a)g(n/b) + f(mn)g(1) \\ &= \sum_{a|m,b|n} f(a)f(b)g(m/a)g(n/b) - f(m)f(n) + f(mn) \\ &= \sum_{a|m} f(a)g(m/a) \sum_{b|n} f(b)g(n/b) - f(m)f(n) + f(mn) \\ &= h(m)h(n) - f(m)f(n) + f(mn) \end{aligned}$$

Now if $h$ is multiplicative, $f(m)f(n) = f(mn)$, which implies $f$ is multiplicative as well. □

**Definition 2.2** (Dirichlet Inverse)**.** *If $f$ is an arithmetic function, we call $g$ **dirichlet inverse** of $f$, denoted by $f^{-1}$, if $f * g = \varepsilon$.*

Recall that $\varepsilon$ is the unit function: $\varepsilon(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$

**Theorem 4.** *If $f$ is an arithmetic function where $f(1) \neq 0$, then $f^{-1}$ exists.*

*Proof.* Let $g = \begin{cases} \frac{1}{f(1)} & \text{if } n = 1 \\ -\frac{\sum_{d|n,d<n} f(n/d)g(d)}{f(1)} & \text{if } n > 1 \end{cases}$
Clearly, $(f * g)(1) = 1$. Now for $n > 1$,

$$\begin{aligned} (f * g)(n) &= (g * f)(n) \\ &= \sum_{d|n} g(d)f(n/d) \\ &= g(n)f(1) + \sum_{d|n,d<n} g(d)f(n/d) \end{aligned}$$

Now,

$$g(n) = -\frac{\sum_{d|n,d<n} g(d)f(n/d)}{f(1)} \implies -g(n)f(1) = \sum_{d|n,d<n} g(d)f(n/d)$$

So, $(f * g)(n) = g(n)f(1) - g(n)f(1) = 0$. Hence, $(f * g) = \varepsilon$, thus $g = f^{-1}$. □

**Corollary 2.** *If $f$ is multiplicative, so is $f^{-1}$*

*Proof.* Since $\varepsilon = f * f^{-1}$, and both $\varepsilon$ and $f$ ar multiplicative, by **theorem 3**, $f^{-1}$ is also multiplicative. □



## 3 More on common multiplicative functions

### 3.1 Number-of-Divisors Function

Let $\tau(n)$ denote the number of divisors of $n$. So, $\tau(n) = \sum_{d|n} 1$. It is easy to see that $\tau$ is multiplicative, since $\tau(n) = \sum_{d|n} 1 = \sum_{d|n} 1(d)1(n/d) = (1 * 1)(n)$. Recall $1(n)$ is the constant function, which is completely multiplicative. Hence, by **theorem 2**, $\tau$ is also multiplicative.
Now it is easy to derive the formula of this function. What is $\tau(p^x)$ where $p$ is prime? Of course the divisors of $p^x$ are $1, p, p^2, \ldots, p^x$, so total (x + 1) divisors. Now for any $n = \prod_{i=1}^{r} p_i^{e_i}$, we can simply apply **theorem 1** and get $\tau(n) = \prod_{i=1}^{r} (e_i + 1)$. $\tau(n)$ can be calculated with a simple loop in $O(\sqrt{n})$.

### 3.2 Sum-of-Divisors Function

The sum of divisors function is denoted by $\sigma$. We can write $\sigma(n) = \sum_{d|n} d$. Similar to the previous function, we can write $\sigma(n) = \sum_{d|n} d = \sum_{d|n} Id(d)1(n/d) = (Id * 1)(n)$. Since $Id$ and 1 are both multiplicative, by **theorem 2**, $\sigma$ is also multiplicative.
Now, note that for a prime $p$, $\sigma(p^x) = 1 + p + p^2 + \cdots + p^x = \frac{p^{x+1}-1}{p-1}$. So, for $n = \prod_{i=1}^{r} p_i^{e_i}$, $\sigma(n) = \prod_{i=1}^{r} \frac{p_i^{e_i+1}-1}{p_i-1}$.

### 3.3 Möbius Function

**Definition 3.1.** *For a positive integer $n$, the **möbius function**, denoted by $\mu$, is defined as,*

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } p^2|n \text{ for some prime } p \\ (-1)^r & \text{if } n = p_1 p_2 \ldots p_r, \text{ where } p_i \text{ are distinct primes} \end{cases}$$

The next theorem shows a very important property of möbius function.

**Theorem 5.** $\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases} = \varepsilon(n)$

*Proof.* For $n = 1$, $\sum_{d|n} \mu(d) = \mu(1) = 1$
When $n > 1$, let $n = \prod_{i=1}^{r} p_i^{e_i}$. If $d|n$, then $d = \prod_{i=1}^{r} p_i^{f_i}$, where $f_i \leq e_i$, for $1 \leq i \leq r$. Now if $\exists i$ such that $f_i > 1$, $\mu(d)$ will be 0. So, the only divisors we care about are the ones with $f_i \leq 1, \forall i$. This means we have $r$ distinct primes, and we will go through each possible combinations and add their contribution to the sum accordingly. For example, if $d$ contains $k$ primes, then we have $\binom{r}{k}$



possible values for $d$, and each of them will contribute $(-1)^k$ to the sum. So,

$$\begin{aligned} \sum_{d|n} \mu(d) &= \sum_{\substack{(f_1,f_2,\ldots,f_r) \\ f_i \in \{0,1\}}} \mu(\prod_{i=1}^{r} p_i^{f_i}) \\ &= (-1)^0 \binom{r}{0} + (-1)^1 \binom{r}{1} + \cdots + (-1)^r \binom{r}{r} \\ &= \sum_{i=0}^{r} \binom{r}{i} (-1)^i \\ &= (1-1)^r \\ &= 0 \end{aligned}$$

□

We also conclude that $\mu$ is **multiplicative** from this theorem, since,

$$\sum_{d|n} \mu(d) = \sum_{d|n} \mu(d)1(n/d) = (\mu * 1)(n) = \varepsilon(n)$$

This implies $\mu$ is the dirichlet inverse of the constant function. So, by **Corollary 2**, $\mu$ is multiplicative. Since $\mu$ is the dirichlet inverse of the constant function, using the definition of $g$ in **theorem 4**, we can write that,

$$\mu(n) = \sum_{d|n,d<n} -\mu(d)$$

Now we can use this in sieve to calculate the value of möbius function for integers from 1 to $n$. Here's an implementation in C++:

```
mu[1] = 1;
for(int i = 1; i < N; i++) {
    for(int j = 2 * i; j < N; j+=i) {
        mu[j] -= mu[i];
    }
}
```

This is basically the standard sieve: instead of iterating over $d$ for each $n$, for each $d$, we loop over its multiples and add its contribution to the answer of the multiples. So, it runs in $O(n \log n)$ time.

### 3.4 Euler's Totient Function

**Definition 3.2.** *For a positive integer $n$, **Euler's Totient Function**, denotes as $\phi(n)$, is defined as the number of positive integers $i$ up to $n$ such that $gcd(i, n) = 1$.*

For example, $\phi(4) = 2$, since from 1 to 4, only 1 and 3 are coprimes with 4.

**Theorem 6.** $\sum_{d|n} \phi(d) = n$

*Proof.* Let $S_d$ be the set of all positive integers up to $n$ whose gcd with $n$ is $d$. Now, $gcd(n, m) = d$ implies $gcd(n/d, m/d) = 1$. By definition, the number of such $m$ is $\phi(n/d)$. So,

$$\sum_{d|n} |S_d| = \sum_{d|n} \phi(n/d)$$

Clearly, the sets $S_d$ for all $d$ are pairwise disjoint. Moreover, for every integer $1 \le k \le n$, there exists a $d$ such that $d|n$ and $k \in S_d$. Hence, $\sum_{d|n} |S_d| = \sum_{d|n} \phi(n/d) = \sum_{d|n} \phi(d) = n$. □

Since, $\sum_{d|n} \phi(d) = \sum_{d|n} \phi(d)1(n/d) = (\phi * 1)(n) = Id(n)$, we also see that $\phi$ is a multiplicative function.
Note that if $p$ is prime, $\phi(p) = p - 1$, and $\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1)$. So, if $n = \prod_{i=1}^{r} p_i^{e_i}$, by **theorem 1**, we get

$$\phi(n) = \prod_{i=1}^{r} p_i^{e_i - 1}(p_i - 1) = \prod_{i=1}^{r} p_i^{e_i} \frac{(p_i - 1)}{p_i} = n \prod_{i=1}^{r} (1 - \frac{1}{p_i})$$

which is the well-known formula for calculating $\phi(n)$.
About implementation, we can simply initialize $\phi(n) = n$, and for each prime divisor $p$, we update its value by multiplying it with $(1 - \frac{1}{p}) = \frac{p-1}{p}$. Here's the code:

```
for (int i = 1; i <= n; i++) phi[i] = i;
for(int p = 2; p <= n; p++) {
  if (phi[p] == p) {
    phi[p] = p - 1;
    for (int i = 2 * p; i <= n; i += p) {
      phi[i] = (phi[i] / p) * (p - 1);
    }
  }
}
```

## 4 The Möbius Inversion

**Theorem 7.** *Let $f$ and $g$ be two arithmetic functions. Then, $g(n) = \sum_{d|n} f(d)$ if and only if $f(n) = \sum_{d|n} g(d)\mu(n/d)$.*

*Proof.* Note that $g(n) = \sum_{d|n} f(d) = \sum_{d|n} f(d)1(n/d) = (f * 1)(n)$, which implies $g = f * 1$.
Similarly, $f(n) = \sum_{d|n} g(d)\mu(n/d) = (g * \mu)(n)$, which implies $f = g * \mu$. Hence it suffices to show that $g = f * 1$ if and only if $f = g * \mu$.

$$g * \mu = (f * 1) * \mu = f * (1 * \mu) = f * \varepsilon = f$$

. Now for converse,

$$f * 1 = (g * \mu) * 1 = g * (1 * \mu) = g * \varepsilon = g$$

. □

I kept this single theorem in a separate section only because it is probably the most useful part of the entire note!

## 5 Problems

**Problem 1.** Given $n$, calculate the number of pairs $(i, j)$ such that $i, j \in [1, n]$ and $gcd(i, j) = 1$.
**Solution.** We basically need to find the value of $\sum_{i=1}^{n} \sum_{j=1} [gcd(i, j) = 1]$. Now note that $[gcd(i, j) = 1] = \varepsilon(gcd(i, j))$. Hence,



$$\sum_{i=1}^{n} \sum_{j=1}^{n} [gcd(i, j) = 1] = \sum_{i=1}^{n} \sum_{j=1}^{n} \varepsilon(gcd(i, j)) = \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{d|gcd(i,j)} \mu(d)$$

Now, $d|gcd(i, j)$ means $d|i$ and $d|j$. So,

$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{d|gcd(i,j)} \mu(d) &= \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{d=1}^{n} [d \mid i][d \mid j]\mu(d) \\ &= \sum_{d=1}^{n} \mu(d) \sum_{i=1}^{n} [d \mid i] \sum_{j=1}^{n} [d \mid j] \\ &= \sum_{d=1}^{n} \mu(d) (\lfloor \frac{n}{d} \rfloor)^2\end{aligned}$$

Note that the number of multiples of $d$ from 1 to $n$ is $\lfloor \frac{n}{d} \rfloor$. We can calculate the value of $\mu(i)$ for integers up to $n$ with sieve, and calculate the final answer with a simple loop.
The sequence $\lfloor \frac{n}{1} \rfloor, \lfloor \frac{n}{2} \rfloor, \ldots, \lfloor \frac{n}{n} \rfloor$ contains at most $2\sqrt{n}$ unique values, so if we precalculate the $\mu(i)$, we can calculate the answer for each $n$ in $O(\sqrt{n})$ time.

```
for (int l = 1, r; l <= n; l = r + 1) {
  r = n / (n / i);
  //now for all l <= i <= r, n / i is equal.
}
```

**Problem 2.** Given $n$, calculate the number of pairs $(i, j)$ such that $1 \le i < j \le n$ and $gcd(i, j) = 1$.
**Solution.** We need to calculate $\sum_{i=1}^{n} \sum_{j=i+1}^{n} [gcd(i, j) = 1]$.

$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=i+1}^{n} [gcd(i, j) = 1] &= \sum_{j=2}^{n} \sum_{i=1}^{j-1} [gcd(i, j) = 1] \\ &= \sum_{j=2}^{n} \phi(j)\end{aligned}$$

Now it's a simple loop!
**Problem 3.** Given $a, b, k$, calculate the number of pairs $(x, y)$ such that $x \in [1, a], y \in [1, b]$ and $gcd(x, y) = k$. Also, $(x, y)$ and $(y, x)$ are considered the same.
**Solution.** First of all, note that $gcd(x, y) = k$ implies that $gcd(x/k, y/k) = 1$. So, we can rephrase the statement: calculate the number of pairs $(x, y)$ such that $x \in [1, \lfloor \frac{a}{k} \rfloor], y \in [1, \lfloor \frac{b}{k} \rfloor]$ and $gcd(x, y) = 1$.
Let $n = \lfloor \frac{a}{k} \rfloor$ and $m = \lfloor \frac{b}{k} \rfloor$. For simplicity, let's assume $n \ge m$ (otherwise we can simply swap $n$, $m$). We need to calculate $\sum_{i=1}^{n} \sum_{j=1}^{min(m,i)} [gcd(i, j) = 1]$.



$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=1}^{min(m,i)} [gcd(i, j) = 1] &= \sum_{i=1}^{m} \sum_{j=1}^{i} [gcd(i, j) = 1] + \sum_{i=m+1}^{n} \sum_{j=1}^{m} [gcd(i, j) = 1] \\ &= \sum_{i=1}^{m} \phi(i) + \sum_{i=m+1}^{n} \sum_{j=1}^{m} \sum_{d|gcd(i,j)} \mu(d) \\ &= \sum_{i=1}^{m} \phi(i) + \sum_{i=m+1}^{n} \sum_{j=1}^{m} \sum_{d|i,d|j} \mu(d) \\ &= \sum_{i=1}^{m} \phi(i) + \sum_{i=m+1}^{n} \sum_{j=1}^{m} \sum_{d=1}^{m} [d \mid i][d \mid j]\mu(d) \\ &= \sum_{i=1}^{m} \phi(i) + \sum_{d=1}^{m} \mu(d) \sum_{i=m+1}^{n} [d \mid i] \sum_{j=1}^{m} [d \mid j] \\ &= \sum_{i=1}^{m} \phi(i) + \sum_{d=1}^{m} \mu(d) (\lfloor \frac{n}{d} \rfloor - \lfloor \frac{m}{d} \rfloor)(\lfloor \frac{m}{d} \rfloor)\end{aligned}$$

Now you can calculate it with a simple loop in $O(m)$.

**Problem 4.** Given $n$, calculate $\sum_{i=1}^{n} \sum_{j=1}^{n} gcd(i, j)$.
**Solution.** From **theorem 6**, we know that $\sum_{d|n} \phi(d) = n$. So,

$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=1}^{n} gcd(i, j) &= \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{d|gcd(i,j)} \phi(d) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{d=1}^{n} [d \mid i][d \mid j]\phi(d) \\ &= \sum_{d=1}^{n} \phi(d) (\sum_{i=1}^{n} [d \mid i])(\sum_{j=1}^{n} [d \mid j]) \\ &= \sum_{d=1}^{n} \phi(d) \lfloor \frac{n}{d} \rfloor^2\end{aligned}$$

which can be calculated in $O(\sqrt{n})$ for each $n$ after the sieve (which would take $O(n \log n)$ or $O(n)$ if you use linear sieve).

**Problem 5.** (Hyperrectangle GCD)Let there be a $k$-dimensional hyperrectangle with dimensions $n_1, n_2, \ldots n_k$. Each unit cell of the hyperrectangle is addressed by $(p_1, p_2, \ldots, p_k)$ where $1 \le p_i \le n_i$ for all $i \le k$. The value of a cell is the **gcd** of its coordinates. Calculate the sum of value over all cells of the hyperrectangle.
**Solution.** We are basically asked to calculate the value of:

$$\sum_{x_1=1}^{n_1} \sum_{x_2=1}^{n_2} \cdots \sum_{x_k=1}^{n_k} gcd(x_1, x_2, \ldots, x_k)$$



Let $cnt[x]$ be the number of such tuples whose gcd is $x$. Also, let $n = \min_{i=1}^{k} n_i$. Then our answer will be $\sum_{g=1}^{n} g \times cnt[g]$. We can write $g = \sum_{d|g} \phi(d)$. So,

$$\sum_{g=1}^{n} g \times cnt[g] = \sum_{g=1}^{n} \sum_{d|g} \phi(d) \times cnt[g]$$

Instead of iterating over divisors $d$ of each $g$, we can iterate over multiples $g$ of each $d$. So,

$$\begin{aligned}\sum_{g=1}^{n} g \times cnt[g] &= \sum_{g=1}^{n} \sum_{d|g} \phi(d) \times cnt[g] \\ &= \sum_{d=1}^{n} \phi(d) \sum_{d|g} cnt[g]\end{aligned}$$

Now, $\sum_{d|g} cnt[g]$ is basically the number of tuples whose gcd is divisible by $d$. It is easy to see that the number of such tuples are $\lfloor \frac{n_1}{d} \rfloor \times \lfloor \frac{n_2}{d} \rfloor \times \cdots \times \lfloor \frac{n_k}{d} \rfloor$. Hence the answer is,

$$\sum_{d=1}^{n} \phi(d) (\lfloor \frac{n_1}{d} \rfloor \times \lfloor \frac{n_2}{d} \rfloor \times \cdots \times \lfloor \frac{n_k}{d} \rfloor)$$

which is calulatable in $O(nk)$.
**Problem 6.** (Devu and Birthday Celebration) Answer $Q$ ($Q \le 10^5$) queries: Given $n$ and $f$ ($n, f \le 10^5$), calculate the number of ways to distribute $n$ sweets among $f$ friends so that if $i$-th friend got $a_i$ sweets, $a_i \ge 1$ for all $i$ and $gcd(a_1, a_2, \ldots, a_f) = 1$. Note that the ordering matters: $a = [1, 2]$ and $a = [2, 1]$ are considered different.
**Solution.** First of all, how many ways are there to distribute $n$ sweets among $f$ people so that everyone gets at least one sweet (ignoring the gcd condition)? For that, we can represent a distribution with a string of **'o'**and **'|'**, where **'o'** means we give a sweet to the current person and **'|'** means we move to the next person. For example, if $n = 6$ and $f = 3$, then **'oo|ooo|o'** denotes $a = \{2, 3, 1\}$. Since the string has $n$ **'o'**s,we have $(n - 1)$ places to insert $(f - 1)$ **'|'**s, which can be done in $\binom{n-1}{f-1}$ ways.
Now for a fixed $f$, let $h(n)$ be the number of ways to distribute $n$ sweets among $f$ people so that everyone gets at least one sweet. From the explanation above, we get $h(n) = \binom{n-1}{f-1}$. Now let $f(n)$ be the number of ways to distribute $n$ sweets among $f$ people so that each gets at least 1 and gcd of all $a_i$'s is 1 (which is basically the answer of the problem).
For a particular distribution, if $gcd(a_1, a_2, \ldots a_f) = d$, then $d|n$.

$$\begin{aligned}&a_1 + a_2 + \cdots + a_f = n \\ \implies &d(a'_1 + a'_2 + \cdots + a'_f) = n \\ \implies &a'_1 + a'_2 + \cdots + a'_f = \frac{n}{d}\end{aligned}$$

where $gcd(a'_1, a'_2, \ldots, a'_f) = 1$. Since $a'_1 + a'_2 + \cdots + a'_f = \frac{n}{d}$ and $gcd(a'_1, a'_2, \ldots, a'_f) = 1$, by definition, the number of such $a'$ is $f(n/d)$. Hence, we get

$$h(n) = \sum_{d|n} f(n/d) = \sum_{d|n} f(d)$$



Finally, we can apply möbius inversion!

$$f(n) = \sum_{d|n} h(d)\mu(n/d) = \sum_{d|n} \binom{d-1}{f-1} \mu(n/d)$$

We can simply precalculate the $\mu(i)$ and the divisors for integer up to $10^5$ to answer the queries efficiently.
**Problem 7.** (COPRIME3) Given an array $a$ of length $n$ ($1 \le a[i] \le 10^6, 1 \le n \le 10^5$), calculate the number of triplets $i < j < k$ such that $gcd(a[i], a[j], a[k]) = 1$.
**Solution.** We basically need to calculate $\sum_{i=1}^{n} \sum_{j=i+1}^{n} \sum_{k=j+1}^{n} [gcd(a[i], a[j], a[k]) = 1]$. So,

$$\begin{aligned}\sum_{i=1}^{n} \sum_{j=i+1}^{n} \sum_{k=j+1}^{n} [gcd(a[i], a[j], a[k]) = 1] &= \sum_{i=1}^{n} \sum_{j=i+1}^{n} \sum_{k=j+1}^{n} \varepsilon(gcd(a[i], a[j], a[k])) \\ &= \sum_{i=1}^{n} \sum_{j=i+1}^{n} \sum_{k=j+1}^{n} \sum_{d|gcd(a[i],a[j],a[k])} \mu(d) \\ &= \sum_{d=1}^{MAXV} \mu(d) \sum_{i=1}^{n} \sum_{j=i+1}^{n} \sum_{k=j+1}^{n} [d|gcd(a[i], a[j], a[k])]\end{aligned}$$

Now, $d|gcd(a[i], a[j], a[k])$ implies $d$ separately divides each of them.
So, $\sum_{i=1}^{n} \sum_{j=i+1}^{n} \sum_{k=j+1}^{n} [d|gcd(a[i], a[j], a[k])]$ is basically the number of triplets whose elements are divisible by $d$. If $cnt[d]$ is the number of elements in the array $a$ divisible by $d$, then number of such triplets will be $\binom{cnt[d]}{3}$. Calculating $cnt[]$ array is pretty easy: if $freq[x]$ is the number of occurrances of $x$, then $cnt[d] = \sum_{d|x} freq[x]$. Since we will be iterating over multiples for each $d$, it will take $O(n \log n)$ time. Hence we get the final answer:

$$\sum_{d=1}^{MAXV} \mu(d) \binom{cnt[d]}{3}$$

**Problem 8.** (CF Mike and Foam) You are given an array $a$ and $q$ queries. ($|a|, q \le 2 \times 10^5, a[i] \le 5 \times 10^5$). You also have a list $L$, which is initially empty. In each query, you will be given an integer $x$. If $x \in L$, then you will remove $x$ from $L$, otherwise add $x$ to $L$. Then you need to output the number of pairs $(i, j)$ where $i < j$ and $i, j \in L$ such that $gcd(a_i, a_j) = 1$.
**Solution.** Exercise. Check my code if needed here.
**Problem 9.** (CF The Holmes Children) Let $f(n)$ be the number of distinct positive integer pairs $(x, y)$ such that $x + y = n$ and $gcd(x, y) = 1$ (additionally, $f(1) = 1$). Let $g(n) = \sum_{d|n} f(d)$. Given $n$ and $k$, calculate $F_k(n)$, which is defined as:

$$F_k(n) = \begin{cases} f(g(n)) & \text{if } k = 1 \\ g(F_{k-1}(n)) & \text{if } k > 1 \text{ and } k \bmod 2 = 0 \\ f(F_{k-1}(n)) & \text{if } k > 1 \text{ and } k \bmod 2 = 1 \end{cases}$$

**Solution.** Exercise.



## References


- Multiplicative Function, Wikipedia
- Beginning Number Theory, Neville Robbins
- [Tutorial] Math note — Möbius inversion

A000079 Powers of 2: a(n) = 2^n. 3328
(Formerly M1129 N0432)

1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536, 131072, 262144, 524288, 1048576, 2097152, 4194304, 8388608, 16777216, 33554432, 67108864, 134217728, 268435456, 536870912, 1073741824, 2147483648, 4294967296, 8589934592

A000110 Bell or exponential numbers: number of ways to partition a set of n labeled elements. 1371
(Formerly M1484 N0585)

1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, 115975, 678570, 4213597, 27644437, 190899322, 1382958545, 10480142147, 82864869804, 682076806159, 5832742205057, 51724158235372, 474869816156751, 4506715738447323, 44152005855084346, 445958869294805289, 4638590332229999353, 49631246523618756274

A000040 The prime numbers. 11567
(Formerly M0652 N0241)

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271

A000010 Euler totient function phi(n): count numbers <= n and prime to n. 4245
(Formerly M0299 N0111)

1, 1, 2, 2, 4, 2, 6, 4, 6, 4, 10, 4, 12, 6, 8, 8, 16, 6, 18, 8, 12, 10, 22, 8, 20, 12, 18, 12, 28, 8, 30, 16, 20, 16, 24, 12, 36, 18, 24, 16, 40, 12, 42, 20, 24, 22, 46, 16, 42, 20, 32, 24, 52, 18, 40, 24, 36, 28, 58, 16, 60, 30, 36, 32, 48, 20, 66, 32, 44

A000142 Factorial numbers: n! = 1*2*3*4*...*n (order of symmetric group S_n, number of permutations of n letters). 2944
(Formerly M1675 N0659)

1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, 39916800, 479001600, 6227020800, 87178291200, 1307674368000, 20922789888000, 355687428096000, 6402373705728000, 121645100408832000, 2432902008176640000, 51090942171709440000, 1124000727777607680000

A000108 Catalan numbers: C(n) = binomial(2n,n)/(n+1) = (2n)!/(n!(n+1)!). 4209
(Formerly M1459 N0577)

1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, 208012, 742900, 2674440, 9694845, 35357670, 129644790, 477638700, 1767263190, 6564120420, 24466267020, 91482563640, 343059613650, 1289904147324, 4861946401452, 18367353072152, 69533550916004, 263747951750360, 1002242216651368, 3814986502092304

A000984 Central binomial coefficients: binomial(2*n,n) = (2*n)!/(n!)^2. 1128
(Formerly M1645 N0643)

1, 2, 6, 20, 70, 252, 924, 3432, 12870, 48620, 184756, 705432, 2704156, 10400600, 40116600, 155117520, 601080390, 2333606220, 9075135300, 35345263800, 137846528820, 538257874440, 2104098963720, 8233430727600, 32247603683100, 126410606437752, 495918532948104, 1946939425648112

A001175 Pisano periods (or Pisano numbers): period of Fibonacci numbers mod n. 177
(Formerly M2710 N1087)

1, 3, 8, 6, 20, 24, 16, 12, 24, 60, 10, 24, 28, 48, 40, 24, 36, 24, 18, 60, 16, 30, 48, 24, 100, 84, 72, 48, 14, 120, 30, 48, 40, 36, 80, 24, 76, 18, 56, 60, 40, 48, 88, 30, 120, 48, 32, 24, 112, 300, 72, 84, 108, 72, 20, 48, 72, 42, 58, 120, 60, 30, 48, 96, 140, 120, 136

A008277 Triangle of Stirling numbers of the second kind, S2(n,k), n >= 1, 1 <= k <= n. 645

1, 1, 1, 1, 3, 1, 1, 1, 7, 6, 1, 1, 15, 25, 10, 1, 1, 31, 90, 65, 15, 1, 1, 63, 301, 350, 140, 21, 1, 1, 127, 966, 1701, 1050, 266, 28, 1, 1, 255, 3025, 7770, 6951, 2646, 462, 36, 1, 1, 511, 9330, 34105, 42525, 22827, 5880, 750, 45, 1, 1, 1023, 28501, 145750, 246730, 179487, 63987, 11880, 1155, 55, 1

(list; table; graph; refs; listen; history; text; internal format)

A000166 Subfactorial or rencontres numbers, or derangements: number of permutations of n elements with no fixed points. 545
(Formerly M1937 N0766)

1, 0, 1, 2, 9, 44, 265, 1854, 14833, 133496, 1334961, 14684570, 176214841, 2290792932, 32071101049, 481066515734, 7697064251745, 130850092279664, 2355301661033953, 44750731559645106, 895014631192902121, 18795307255050944540, 413496759611120779881, 9510425471055777937262

$$!n = n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}$$

A000041 a(n) is the number of partitions of n (the partition numbers). 3896
(Formerly M0663 N0244)

1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77, 101, 135, 176, 231, 297, 385, 490, 627, 792, 1002, 1255, 1575, 1958, 2436, 3010, 3718, 4565, 5604, 6842, 8349, 10143, 12310, 14883, 17977, 21637, 26015, 31185, 37338, 44583, 53174, 63261, 75175, 89134, 105558, 124754, 147273, 173525

## A tutorial of reflection principle in combinatorics

By -firefly-, history, 11 months ago,

### Introduction

A classical problem reads:

> *Count the paths from $(0,0)$ to $(m,n)(n>m>0)$ that doesn't go through $y=x$ (the path can touch the line). One can only move up or right by $1$ in one move.*

The problem can also be described as:

> *Count binary strings that consist of $m$ zeros and $n$ ones $(n \geq m)$, such that in every prefix of the string, the number of zeros doesn't exceed the number of ones.*

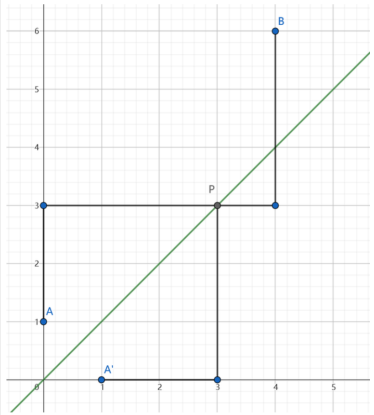
Consider how to count paths from $(0,0)$ to $(m,n)$ without the intersection restriction. To get to $(m,n)$, we must move up $n$ times and move right $m$ times. In other words, we need to permute $n$ right and $m$ up, whose quantity is

$$\frac{(m+n)!}{m!n!} = \binom{m+n}{m}$$

To find the answer, we can try to count the paths that go through $y=x$, and this is when the reflection principle is utilized. The principle says:

> *Given point $A, B$ at the same side of line $l$, the number of paths from $A$ to $B$ that intersect with $l$ is equal to that of paths from $A'$ to $B$, where $A'$ is the reflection point through $l$.*

To show why it is correct, we should first note that a path from $A'$ to $B$ must intersect with $y=x$. Say at point $P$ the path first intersect with $y=x$, we can reflect the path from $A'$ to $P$ through $l$ and get a path from $A$ to $B$, and vice versa. This indicates that the intersecting paths from $A$ to $B$ and the paths from $A'$ and $B$ are one-to-one corresponded, and the quantity of the two are the same.




Back to our problem. Going through $y=x$ is the same as intersecting with $y=x-1$ in this case, so $(0,0)$ can reflected to $(1,-1)$, and the number of intersecting path is $\binom{m+n}{m-1}$. Therefore, the answer to the problem is

$$\boxed{\binom{m+n}{m} - \binom{m+n}{m-1} = \frac{n-m+1}{n+1}\binom{m+n}{m}}$$

In particular, when $n=m$, the formula becomes

$$\boxed{\frac{1}{n+1}\binom{2n}{n}}$$

which is the **Catalan number**.

Note

Note that there are two ways to generate valid paths in the first description when $n=m$, one keeps $x \geq y$ and the other keeps $x \leq y$, so the answer is $\frac{2}{n+1}\binom{2n}{n}$.

Sprague-Grundy theorem

Let's consider a state $v$ of a two-player impartial game and let $v_i$ be the states reachable from it (where $i \in \{1,2,\ldots,k\}, k \geq 0$). To this state, we can assign a fully equivalent game of Nim with one pile of size $x$. The number $x$ is called the Grundy value or nim-value of state $v$.

Moreover, this number can be found in the following recursive way:

$$x = \operatorname{mex}\{x_1, \ldots, x_k\},$$

where $x_i$ is the Grundy value for state $v_i$ and the function $\operatorname{mex}$ (*minimum excludant*) is the smallest non-negative integer not found in the given set.

Viewing the game as a graph, we can gradually calculate the Grundy values starting from vertices without outgoing edges. Grundy value being equal to zero means a state is losing.

## Chicken McNugget Theorem

The **Chicken McNugget Theorem** (or **Postage Stamp Problem** or **Frobenius Coin Problem**) states that for any two relatively prime positive integers $m, n$, the greatest integer that cannot be written in the form $am + bn$ for nonnegative integers $a, b$ is $mn - m - n$.

A consequence of the theorem is that there are exactly $\frac{(m-1)(n-1)}{2}$ positive integers which cannot be expressed in the form $am + bn$. The proof is based on the fact that in each pair of the form $(k, mn-m-n-k)$, exactly one element is expressible.

### 2. Distinct values of $\left\lfloor \frac{n}{i} \right\rfloor$

Now, let's get to even more interesting facts:

Fact 2

Over all positive integers $i$, there are $O(\sqrt{n})$ distinct values of $\left\lfloor \frac{n}{i} \right\rfloor$.

We will discuss why this is true and how to retrieve the exact values over the exact ranges in $O(\sqrt{n})$.

Proof of Fact 2

We note that for $i > \sqrt{n}$, we have that $\left\lfloor \frac{n}{i} \right\rfloor \leq \frac{n}{i} < \frac{n}{\sqrt{n}} = \sqrt{n}$ and there are obviously at most $\sqrt{n}$ values for $i \leq \sqrt{n}$, this makes a total of at most $2\sqrt{n} \in O(\sqrt{n})$.

Now, note that these values are non-increasing as $i$ increases. For example, for $n = 10$, we get the values (starting from $i = 1$) $10, 5, 3, 2, 2, 1, 1, 1, 1, 1, 0, 0, \ldots$. So, to find these values, we will prove the following fact:

Fact 3

Let $S = \left\{ \left\lfloor \frac{n}{i} \right\rfloor : i \in \mathbb{Z}^+ \right\}$ — in other words, the set of all distinct values of $\left\lfloor \frac{n}{i} \right\rfloor$. Let $q \in S$ where $q > 0$ and let $l$ be the minimum positive integer such that $\left\lfloor \frac{n}{l} \right\rfloor = q$ and let $r$ be the maximum positive integer such that $\left\lfloor \frac{n}{r} \right\rfloor = q$, then

$$r = \left\lfloor \frac{n}{\left\lfloor \frac{n}{l} \right\rfloor} \right\rfloor$$

## Rearrangement Inequality

The **Rearrangement Inequality** states that, if $A = \{a_1, a_2, \cdots, a_n\}$ is a permutation of a finite set (in fact, multiset) of real numbers and $B = \{b_1, b_2, \cdots, b_n\}$ is a permutation of another finite set of real numbers, the quantity $a_1b_1 + a_2b_2 + \cdots + a_nb_n$ is maximized when $A$ and $B$ are similarly sorted (that is, if $a_k$ is greater than or equal to exactly $i$ of the other members of $A$, then $b_k$ is also greater than or equal to exactly $i$ of the other members of $B$). Conversely, $a_1b_1 + a_2b_2 + \cdots + a_nb_n$ is minimized when $A$ and $B$ are oppositely sorted (that is, if $a_k$ is less than or equal to exactly $i$ of the other members of $A$, then $b_k$ is also greater than or equal to exactly $i$ of the other members $B$).

## Wilson's Theorem

In number theory, **Wilson's Theorem** states that if integer $p > 1$, then $(p-1)! + 1$ is divisible by $p$ if and only if $p$ is prime. It was stated by John Wilson. The French mathematician Lagrange proved it in 1771.

# Primitive Root

## Definition

In modular arithmetic, a number $g$ is called a `primitive root modulo n` if every number coprime to $n$ is congruent to a power of $g$ modulo $n$. Mathematically, $g$ is a `primitive root modulo n` if and only if for any integer $a$ such that $\gcd(a, n) = 1$, there exists an integer $k$ such that:

$$g^k \equiv a \pmod{n}.$$

$k$ is then called the `index` or `discrete logarithm` of $a$ to the base $g$ modulo $n$. $g$ is also called the `generator` of the multiplicative group of integers modulo $n$.

In particular, for the case where $n$ is a prime, the powers of primitive root runs through all numbers from $1$ to $n - 1$.

## Existence

Primitive root modulo $n$ exists if and only if:

- $n$ is 1, 2, 4, or
- $n$ is power of an odd prime number ($n = p^k$), or
- $n$ is twice power of an odd prime number ($n = 2 \cdot p^k$).

This theorem was proved by Gauss in 1801.

# Multiplicative order

12 languages

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From Wikipedia, the free encyclopedia

In number theory, given a positive integer *n* and an integer *a* coprime to *n*, the **multiplicative order** of *a* modulo *n* is the smallest positive integer *k* such that $a^k \equiv 1 \pmod{n}$.[1]

In other words, the multiplicative order of *a* modulo *n* is the order of *a* in the multiplicative group of the units in the ring of the integers modulo *n*.

The order of *a* modulo *n* is sometimes written as $\operatorname{ord}_n(a)$.[2]

# Carmichael function

14 languages

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From Wikipedia, the free encyclopedia

In number theory, a branch of mathematics, the **Carmichael function** $\lambda(n)$ of a positive integer $n$ is the smallest positive integer $m$ such that

$$a^m \equiv 1 \pmod{n}$$

holds for every integer $a$ coprime to $n$. In algebraic terms, $\lambda(n)$ is the exponent of the multiplicative group of integers modulo $n$. As this is a finite abelian group, there must exist an element whose order equals the exponent, $\lambda(n)$. Such an element is called a **primitive λ-root modulo *n***.

### Recurrence for $\lambda(n)$ [edit]

The Carmichael lambda function of a prime power can be expressed in terms of the Euler totient. Any number that is not 1 or a prime power can be written uniquely as the product of distinct prime powers, in which case $\lambda$ of the product is the least common multiple of the $\lambda$ of the prime power factors. Specifically, $\lambda(n)$ is given by the recurrence

$$\lambda(n) = \begin{cases} \varphi(n) & \text{if } n \text{ is } 1, 2, 4, \text{ or an odd prime power,} \\ \frac{1}{2}\varphi(n) & \text{if } n = 2^r,\ r \geq 3, \\ \operatorname{lcm}\left(\lambda(n_1), \lambda(n_2), \ldots, \lambda(n_k)\right) & \text{if } n = n_1 n_2 \ldots n_k \text{ where } n_1, n_2, \ldots, n_k \text{ are powers of distinct primes.} \end{cases}$$

Euler's totient for a prime power, that is, a number $p^r$ with $p$ prime and $r \geq 1$, is given by

$$\varphi(p^r) = p^{r-1}(p - 1).$$

# Discrete Logarithm ¶

The discrete logarithm is an integer $x$ satisfying the equation

$$a^x \equiv b \pmod{m}$$

for given integers $a$, $b$ and $m$.

The discrete logarithm does not always exist, for instance there is no solution to $2^x \equiv 3 \pmod{7}$. There is no simple condition to determine if the discrete logarithm exists.

In this article, we describe the **Baby-step giant-step** algorithm, an algorithm to compute the discrete logarithm proposed by Shanks in 1971, which has the time complexity $O(\sqrt{m})$. This is a **meet-in-the-middle** algorithm because it uses the technique of separating tasks in half.

# Discrete Root

The problem of finding a discrete root is defined as follows. Given a prime $n$ and two integers $a$ and $k$, find all $x$ for which:

$$x^k \equiv a \pmod{n}$$

## The algorithm

We will solve this problem by reducing it to the discrete logarithm problem.

Let's apply the concept of a primitive root modulo $n$. Let $g$ be a primitive root modulo $n$. Note that since $n$ is prime, it must exist, and it can be found in $O(Ans \cdot \log\phi(n) \cdot \log n) = O(Ans \cdot \log^2 n)$ plus time of factoring $\phi(n)$.

We can easily discard the case where $a = 0$. In this case, obviously there is only one answer: $x = 0$.

Since we know that $n$ is a prime and any number between 1 and $n - 1$ can be represented as a power of the primitive root, we can represent the discrete root problem as follows:

$$(g^y)^k \equiv a \pmod{n}$$

where

$$x \equiv g^y \pmod{n}$$

This, in turn, can be rewritten as

$$(g^k)^y \equiv a \pmod{n}$$

Now we have one unknown $y$, which is a discrete logarithm problem. The solution can be found using Shanks' baby-step giant-step algorithm in $O(\sqrt{n}\log n)$ (or we can verify that there are no solutions).

Having found one solution $y_0$, one of solutions of discrete root problem will be $x_0 = g^{y_0} \pmod{n}$.

# Permutation Order

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

**Time limit:** 1.00 s **Memory limit:** 512 MB

Let $p(n, k)$ denote the $k$th permutation (in lexicographical order) of $1 \ldots n$. For example, $p(4, 1) = [1, 2, 3, 4]$ and $p(4, 2) = [1, 2, 4, 3]$.

Your task is to process two types of tests:

1. Given $n$ and $k$, find $p(n, k)$
2. Given $n$ and $p(n, k)$, find $k$

**Input**

The first line has an integer $t$: the number of tests.

Each test is either "1 $n$ $k$" or "2 $n$ $p(n, k)$".

**Output**

For each test, print the answer according to the example.

**Constraints**

- $1 \leq t \leq 1000$
- $1 \leq n \leq 20$
- $1 \leq k \leq n!$

```cpp
int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    int t; cin >> t;
    while(t--){
        int c; cin >> c;
        if(c == 1){
            int n; ll k; cin >> n >> k;
            --k;
            vector<int> temp = decimal_to_factoradix(k);
            vector<int> ans = factoradix_to_permutation(n, temp);
            for(auto &x: ans)cout << x << " ";
            cout << nl;
        } else {
            int n; cin >> n;
            vi permutation(n); for(int i=0;i<n;++i) cin >> permutation[i];
            vi factoradix = permutation_to_factoradix(n, permutation);
            ll k = factoradix_to_decimal(factoradix);
            ++k;
            cout << k << nl;
        }
    }
}
```

# Distinct Substrings

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

**Time limit:** 1.00 s **Memory limit:** 512 MB

Count the number of distinct substrings that appear in a string.

**Input**

The only input line has a string of length $n$ that consists of characters a–z.

**Output**

Print one integer: the number of substrings.

**Constraints**

- $1 \leq n \leq 10^5$

**Example**

Input:
`abaa`

Output:
8

Explanation: the substrings are `a`, `b`, `aa`, `ab`, `ba`, `aba`, `baa` and `abaa`.

```cpp
int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    string s; cin >> s;
    sa_init();
    for(auto &c:s)sa_extend(c);

    ll ans = 0;
    for(int i=1;i<sz;++i){
        ans += st[i].len - st[st[i].link].len;
    }
    cout << ans << nl;
}
```

## Distinct Subsequences

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a string. You can remove any number of characters from it, but you cannot change the order of the remaining characters.

How many different strings can you generate?

**Input**

The first input line contains a string of size $n$. Each character is one of a–z.

**Output**

Print one integer: the number of strings modulo $10^9 + 7$.

**Constraints**

- $1 \le n \le 5 \cdot 10^5$

```
int last[26];
ll dp[N];

int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    string s; cin >> s;
    int n = (int)s.length();
    s = '#' + s;

    dp[0] = 1;
    for(int i=1;i<=n;++i){

        dp[i] = dp[i-1]*2;
        if(dp[i] > MOD)dp[i] -= MOD;
        if(last[s[i]-'a'] > 0){
            dp[i] = (dp[i] + MOD - dp[last[s[i]-'a']-1])%MOD;
        }

        last[s[i]-'a'] = i;
    }

    cout << (dp[n]+MOD-1)%MOD << nl;
}
```

## Repeating Substring

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

**Time limit:** 1.00 s **Memory limit:** 512 MB

A repeating substring is a substring that occurs in two (or more) locations in the string. Your task is to find the longest repeating substring in a given string.

**Input**

The only input line has a string of length $n$ that consists of characters a–z.

**Output**

Print the longest repeating substring. If there are several possibilities, you can print any of them. If there is no repeating substring, print $-1$.

**Constraints**

- $1 \le n \le 10^5$

```
int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    sa_init();
    string s; cin >> s;
    for(auto &c:s)sa_extend(c);

    int v = 0, last = 0, tin = 0, stop_at = 0;
    for(auto &c:s){
        v = sa_go(v, c);
        sa_walk(v);

        if(last < mx_len){
            stop_at = tin+1;
        }
        last = mx_len;
        ++tin;
    }

    if(mx_len == 0){
        cout << -1 << nl; return 0;
    }

    for(int i=stop_at-mx_len;i<stop_at;++i)cout << s[i];
    cout << nl;
}
```

## Substring Order I

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a string of length $n$. If all of its distinct substrings are ordered lexicographically, what is the $k$th smallest of them?

**Input**

The first input line has a string of length $n$ that consists of characters a–z.

The second input line has an integer $k$.

**Output**

Print the $k$th smallest distinct substring in lexicographical order.

**Constraints**

- $1 \le n \le 10^5$
- $1 \le k \le \frac{n(n+1)}{2}$
- It is guaranteed that $k$ does not exceed the number of distinct substrings.

```
ll k;

bool vst[MAXLEN*2];
ll dp[MAXLEN*2];
//graph is DAG

string temp="", ans;
bool ok = 0;

ll cum = 0;
void walk(int v){
    if(ok)return;
    if(cum == k){
        ans = temp;
        ok = 1;
        return;
    }

    char c = 'a';
    for(int i=0;i<26;++i){
        int u = st[v].next[(int)c];
        if(u != -1){

            if(dp[u] + 1 + cum < k){
                cum += dp[u] + 1;
            } else {
                cum++;
                temp.pb(c);
                walk(u);
                break;
            }
        }
        ++c;
    }
}

void dfs(int v){
    vst[v] = 1;

    int c = 'a';
    for(int i=0;i<26;++i){
        int u = st[v].next[(int)c];
        if(u != -1 and !vst[u]){
            dfs(u);
        }
        ++c;
    }

    c = 'a';
    for(int i=0;i<26;++i){
        int u = st[v].next[(int)c];
        if(u != -1){
            dp[v] += dp[u] + 1;
        }
        ++c;
    }
}

int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    string s; cin >> s;
    cin >> k;

    sa_init();
    for(auto &c:s)sa_extend(c);

    dfs(0);
    walk(0);
    cout << ans << nl;
}
```

# Substring Order II

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

**Time limit:** 1.00 s **Memory limit:** 512 MB

You are given a string of length $n$. If all of its substrings (not necessarily distinct) are ordered lexicographically, what is the $k$th smallest of them?

**Input**

The first input line has a string of length $n$ that consists of characters a–z.

The second input line has an integer $k$.

**Output**

Print the $k$th smallest substring in lexicographical order.

**Constraints**

- $1 \le n \le 10^5$
- $1 \le k \le \frac{n(n+1)}{2}$

```
vi bucket[MAXLEN*2];
bool vst[MAXLEN*2];
ll dp[MAXLEN*2];

void cal_dp(int v){
    vst[v] = 1;
    for(auto &[c, u]:st[v].next){
        if(!vst[u]){
            cal_dp(u);
        }
    }

    for(auto &[c, u]:st[v].next){
        dp[v] += st[u].occ + dp[u];
    }
}

string ans, temp = "";
bool ok = 0;
ll cum = 0;

void walk(int v){
    if(ok)return;

    for(auto &[c, u]:st[v].next){
        if(cum + st[u].occ + dp[u] < k){
            cum += st[u].occ + dp[u];
        } else {

            if(cum + st[u].occ < k){
                cum += st[u].occ;
                temp.pb(c);
                walk(u);
                return;
            } else { //it's this one then
                temp.pb(c);
                ans = temp;
                ok = 1;
                return;
            }
        }
    }
}

int main(){
    ios::sync_with_stdio(false);cin.tie(nullptr);
    string s; cin >> s;
    int n = (int)s.length();
    cin >> k;
    sa_init();
    for(auto &c:s)sa_extend(c);

    for(int i=0;i<sz;++i){
        bucket[ st[i].len ].pb(i);
    }

    for(int i=n;i>=1;--i){
        for(auto &u:bucket[i]){
            st[st[u].link].occ += st[u].occ;
        }
    }

    cal_dp(0);
    walk(0);

    cout << ans << nl;
}
```

# Techniques (A)

techniques.txt 159 lines

```
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Divide and conquer
  Finding interesting points in N log N
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  Master theorem
  Amortized time complexity
Greedy algorithm
  Scheduling
  Max contiguous subvector sum
  Invariants
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  Vertex coloring
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  Monotone queues / monotone stacks / sliding queues
  Sliding queue using 2 stacks
  Persistent segment tree
```